

BEIRUT ARAB UNIVERSITY  
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# **A Hybrid Semantic Web and Deep Learning Approach for Skincare Recommendation in Lebanon**

*An availability-aware and explainable recommender grounded  
in the European cosmetic ingredient register*

by  
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A thesis submitted in partial fulfilment of the requirements  
for the degree of

**Master of Science in Computer Science**

Supervised by  
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2026



# Declaration

**[TODO: Replace with the exact wording your faculty requires.]**

I declare that this thesis is my own work and that all material taken from other sources has been acknowledged. The dataset, ontology and code described here were produced by the author and are publicly available at the addresses given in Chapter 9.

Lynn El Arwadi

Date

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# Abstract

**[TODO: Write the abstract last, once Chapters 4–8 are final.]**

[note: Numbers that belong here: 12,629 products, 1,463 brands, the share of ingredient mentions resolved against the EU register, and the share of suitability claims that come from a source that stated them.]

**Keywords:** skincare recommendation; semantic web; ontology engineering; knowledge graph; product availability; provenance; Lebanon.



# Acknowledgements

[TODO: Write this whenever you like. It is the one page nobody will ask you to justify.]

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Part I

# Foundations



## Chapter 1

# Introduction

### 1.1 Motivation

A recommendation that cannot be acted on is not a recommendation. This is the ordinary condition of skincare advice in Lebanon. Online skincare platforms, review aggregators and the recommender systems built on top of them are designed around international catalogues, and they assume without stating it that a product named to a user can then be obtained by that user. In a market that imports almost all of what it sells, and whose currency and supply chains have both been unstable for several years, that assumption fails often enough to make the advice unusable.

The scale of the disruption is documented. Lebanon imports approximately 95 % of its pharmaceutical products [58], and the same import dependence applies to the dermocosmetic products sold alongside them in pharmacies. Between 2019 and 2023 the Lebanese pound lost the greater part of its value [91], subsidies were withdrawn, and shortages became routine across product classes [32, 37]. The practical consequence for a consumer is that finding a product is a separate problem from choosing one: patients report searching multiple pharmacies for a single item, to the point that a digital platform was proposed specifically to tell people which pharmacy currently holds which product [17].

Skincare inherits this condition and adds one of its own. A skincare product is not a neutral consumer good. It is applied to skin, it is governed by a legal framework that restricts what may be in it, and a wrong choice has physical consequences that range from a wasted purchase to a sensitisation reaction. Within the European Union, Regulation (EC) No 1223/2009 governs which ingredients may be used, in what concentration, and which must be declared by name on the label [40]. A recommender that cannot say *why* it recommended a product cannot be checked against that framework by a pharmacist, cannot be audited by a regulator, and cannot be corrected when it is wrong.

These two conditions, local availability and regulatory groundedness, are not addressed by the same techniques that made general-purpose recommendation successful. Collaborative and neural methods learn from what users did, and their accuracy depends on having enough of that behaviour to learn from. No suitable Lebanese user–item interaction log was available to this study, and none is likely to become available before a system is deployed to produce one. What does exist, or can be made to exist, is knowledge about the products themselves: what is in them, what the law says about those ingredients, which shop currently sells them, and how much each of those statements can be trusted.

## 1.2 Problem statement

This thesis addresses the construction of a skincare recommender for a market in which product availability cannot be assumed and in which a recommendation must be able to state its reason. Three specific gaps motivate the work, and each is established in detail in Chapter 3.

First, **none of the cosmetic product resources reviewed in Chapter 3 links its *ingredient* layer to a regulatory register**. Ingredients are represented, in several of them, as ontology values or identifiers local to the dataset, so no statement about the regulatory status of a substance is expressible in the model. One system, Jannatin and Ratnasari [59], records a product-level registration identifier, which attests that a product is registered but says nothing about which of its constituents is restricted, under which provision, or at what limit; that precedent is credited in Section 3.3.1 and the gap is stated at ingredient level accordingly. The knowledge graphs that do connect cosmetic ingredients to a regulatory framework contain substances rather than marketed products.

Second, **no reviewed system records who asserted a claim**. When a product record states that a cream suits dry skin, that statement may come from the manufacturer, from a retailer’s marketing copy, or from an inference drawn by whoever built the dataset. These are not equally reliable, and treating them as equally reliable is a modelling decision made by omission rather than on purpose.

Third, **availability is represented in the reviewed literature only at country level**. Abesová et al. [8] model which countries a retailer serves and return a local link, which is a genuine precedent and is credited as one. What no reviewed system represents is whether an individual product is listed by a named seller, at what price, and on what date that was observed. In a market where

supply is the binding constraint, that is the level at which availability has to enter the model, and Section 4.9.2 reports how far the present dataset reaches it.

## 1.3 Research questions

- RQ1** How can an ontology and a knowledge graph be constructed so that skin-care recommendations can be generated together with an explanation of why each product is suitable?
- RQ2** How can product descriptions and user-generated text be used to enrich a skincare knowledge graph, and what proportion of the knowledge so derived survives validation against an authoritative source?
- RQ3** How can ingredient-based similarity be used to compare locally available products with established international products, so that an unobtainable recommendation can be substituted rather than withdrawn?
- RQ4** How can market availability and the strength of the evidence behind each claim be represented so that a recommendation can be filtered, explained and audited?

[note: RQ4 is new relative to the proposal. It names what actually distinguishes the work; without it, availability and provenance appear in the contributions without ever having been asked as questions.]

## 1.4 Contributions

The contributions are ordered by how securely they can be defended, rather than by prominence.

1. **A skincare product dataset for a national market.** 12,629 products from 1,463 brands across 22 product types, combining an international catalogue, Lebanese retail listings and Lebanese manufacturers. Of 295,991 ingredient mentions, 285,917 (96.6%) resolve against the European Commission’s CosIng register, and 9,613 products name an ingredient the register records a restriction against. The per-ingredient mapping to a register identifier is not yet exported with the release, so a restriction is currently traceable through the pipeline rather than from the published file alone (Chapter 4).
2. **An evidence model in which provenance is first class.** *Delivered in the dataset:* every suitability claim carries the tier of the source that asserted

it, and 86.7% of claims sit in the three tiers naming an identifiable source, the remainder being inferred from the formula and separable by query (Section 4.9.1). *Planned*: the ontology that expresses this model in PROV-O terms, built with a named construction methodology and published as a resolvable artefact (Chapter 5).

3. **A knowledge graph populated declaratively and enriched from text**, with the mapping published as an artefact and every proposed term validated against an authoritative source before entering the graph. *Planned* (Chapter 6).
4. **Reasoning-based recommendation with retrieval of ingredient-similar alternatives**, so that when a suitable product is not locally listed, an ingredient-similar product that is listed can be offered alongside it, with the shared composition as the explanation. Similar composition does not establish equivalent concentration, vehicle or tolerability, so these are alternatives rather than substitutes. *Planned* (Chapter 7).
5. **An evaluation reported against a named method**, combining structural, functional, logical and comparative assessment. *Planned* (Chapter 8).

*The claim in one sentence.* The reviewed systems model what a product is. This work additionally models whether a named seller lists it in Lebanon, what it costs there, and the strength of the evidence behind every suitability claim. The date on which a local price was observed is the part of that claim the present release does not yet support.

## 1.5 Scope and limitations

The boundaries are stated here so that they read as decisions rather than as omissions discovered later.

- **Skincare, not cosmetics generally.** Makeup, haircare and fragrance are excluded, and the exclusions are enumerated and published. The dataset is predominantly facial (10,704 products) but also contains 1,925 products for the body, hands, lips and eye area. Evaluation is reported on a face-focused subset, defined in Chapter 8, rather than the non-facial records being discarded to fit a scope sentence.
- **Retail selection, not formulation.** The system helps a consumer choose a product; it does not help a chemist design one.

- **No clinical diagnosis.** Skin concerns are represented as consumer-facing categories aligned to clinical vocabulary where possible, not as diagnoses.
- **No user interaction data.** The dataset is a product resource, not an interaction log. This determines which methods are available and is treated throughout as a scope condition rather than a deficiency.
- **A snapshot, not a live feed.** Lebanese prices move quickly. An observation date is recorded for catalogue prices but for almost no local ones, so the present release records that a seller listed a product at a price without recording when (Section 4.9.2). Local price freshness is a limitation of this release, not a delivered property.

## 1.6 Structure of this thesis

The thesis is organised in three parts.

**Part I** establishes the problem and what is known. Chapter 2 introduces the semantic web technologies, the recommender system concepts, the skin-typing vocabulary and the regulatory framework that the later chapters assume. Chapter 3 reviews the literature, with each section closing by naming the requirement that remains unmet.

**Part II** presents what was built. Chapter 4 describes the dataset and its validation. Chapter 5 describes the ontology, its construction methodology and its published form. Chapter 6 describes the population of the knowledge graph and its enrichment. Chapter 7 describes reasoning-based recommendation and ingredient similarity.

**Part III** assesses the result. Chapter 8 reports the evaluation, and Chapter 9 answers the research questions, states the limitations and identifies what the resource enables next.



## Chapter 2

# Background

This chapter introduces the technologies, concepts and domain facts that the remainder of the thesis assumes. It is deliberately descriptive: the choices made among these options are argued in Chapter 3 and Chapter 5, not here.

## 2.1 Semantic web technologies

### 2.1.1 RDF, RDFS and OWL

The Resource Description Framework represents information as *triples*: a subject, a predicate and an object. Every subject and predicate is identified by an IRI, so a statement made in one dataset can refer to an entity defined in another without coordination between their authors. This property is what makes external grounding possible at all, and it is the mechanism by which an ingredient in this work is connected to its entry in the European register.

RDF Schema adds a light vocabulary for describing classes and properties: RDFS:SUBCLASSOF, RDFS:DOMAIN and RDFS:RANGE. The Web Ontology Language adds the expressiveness needed to state that two classes are disjoint, that a property is functional, or that membership of a class follows from a condition rather than from an assertion. The last of these, the *defined class*, is used throughout Chapter 5 and Chapter 7.

*Defined class, and what it cannot do under open-world semantics.* A class whose membership is stated as an equivalence axiom rather than asserted product by product, so the reasoner decides which products belong and recomputes that decision whenever the data changes.

The obvious formulation does not work here. Writing  $P \equiv \text{Product} \sqcap \neg \exists \text{contains.DeclarableAllergen}$  does not classify a product merely because no allergen triple is present: under the open-world assumption of OWL [103], the absence of a statement is not a statement of absence, and a product whose

formula was never recorded is indistinguishable from one that contains nothing declarable.

Two things follow for this thesis. The membership condition must be stated against an explicit completeness assumption, namely that the formula held for that product is complete as published, recorded per product rather than assumed globally. Recording that property does not by itself change the semantics: a flag asserting completeness is just another triple, and the reasoner still will not conclude absence from it. Closing the world over that product's ingredient relation requires either an explicit closure axiom, a cardinality restriction or an enumeration of what it does contain, or a screening step performed outside the reasoner whose result is then asserted. Chapter 5 must state which of those it uses; the declaration alone is a precondition, not a mechanism. And the class is named for what it detects rather than for a conclusion: `NoListedAllergenDetected`, qualified by the version of the allergen list used, not *SensitiveSafeProduct*. A complete declaration without the listed substances does not establish that a product is safe for every person with sensitive skin, and a class name that says so would assert something the axiom does not support.

## 2.1.2 Description logics and reasoning

OWL is grounded in description logics, a family of decidable fragments of first-order logic. A reasoner performs three tasks that this work relies on. *Classification* computes the full subclass hierarchy implied by the axioms, which is what places products into defined classes. *Consistency checking* determines whether the axioms can be satisfied at all, which turns a contradiction in the data into an error rather than into a plausible wrong answer. *Instance checking* determines whether a given individual belongs to a given class, which is the query that produces a recommendation in Chapter 7.

The distinction between what is *asserted* and what is *inferred* matters for the evidence model of this thesis, and is maintained explicitly: inferred statements are held in a separate named graph so that a consumer of the data can choose whether to accept them.

## 2.1.3 SPARQL

SPARQL is the query language for RDF. A query matches a graph pattern and returns bindings, and it can restrict matching to a named graph, which is how provenance-aware querying is implemented here. Its relevance to this thesis is specific: the functional evaluation in Chapter 8 consists of answering each com-

petency question with a SPARQL query, so the query language is not only an implementation detail but part of the evaluation instrument.

### 2.1.4 SHACL and validation

Reasoning and validation are often confused and do different work. A reasoner derives what *follows* from the stated axioms under the open-world assumption, and the absence of a statement means only that it was not stated. The Shapes Constraint Language checks that a graph *conforms* to stated constraints under a closed-world reading: that every product has exactly one price per offer, that an ingredient count agrees with the formula, that no derived attribute is set on a product with no formula. Both are used in Chapter 6, and neither substitutes for the other.

### 2.1.5 Provenance and controlled vocabularies

Two W3C vocabularies are used in this work and are introduced here neutrally.

PROV-O provides a standard way of stating where a piece of information came from. Three of its terms carry most of the weight: an entity PROV:WASDERIVEDFROM another entity, PROV:WASATTRIBUTEDTO an agent, and was PROV:GENERATEDATTIME a moment. Expressing a locally-defined evidence scale in these terms is what allows another researcher’s provenance-aware tooling to interpret it without reading this document first.

SKOS provides a standard way of expressing a controlled list as concepts rather than strings, with SKOS:PREFLABEL and SKOS:ALTLABEL for labels in more than one language and SKOS:BROADER for hierarchy. Both properties matter in this setting: a query for moisturisers should return night moisturisers without the list being hard-coded, and a Lebanese dataset has a real requirement for Arabic labels.

## 2.2 Recommender systems

### 2.2.1 The three families

Recommender systems are conventionally divided into three families by what they consume as input.

*Collaborative filtering* infers preference from the behaviour of similar users and requires a user–item interaction matrix. *Content-based* methods describe items

by their attributes and match those to a user profile, requiring item descriptions but not necessarily other users. *Knowledge-based* methods reason over explicit domain knowledge and constraints, requiring neither interactions nor a history, and are the family this thesis belongs to. *Hybrid* systems combine two or more.

The families are not equally available in every setting. The determining question is not which performs best in general, but which has its input condition satisfied.

### 2.2.2 Cold start, sparsity and explainability

*The three conditions that shape this work.* **Cold start.** A new user or item has no interaction history, so a collaborative method has nothing to compute from. In a market with no deployed system, *every* user is a cold-start user and every item a cold-start item.

**Sparsity.** Even where interactions exist they cover a small fraction of the user–item space. In this dataset the rating field is populated for roughly half the catalogue and no user history exists at all.

**Explainability.** A recommendation that cannot state its reason cannot be checked. In a domain with physical risk and a statutory framework, this is not a presentational preference but a condition of use.

### 2.2.3 Evaluation

Recommender systems are conventionally evaluated by accuracy or ranking measures computed against held-out interactions, supplemented by diversity, novelty and coverage.

The absence of interaction data constrains that design without eliminating quantitative evaluation. Accuracy and ranking measures computed against observed behaviour do require interactions, and so does novelty defined relative to what a user has seen. Catalogue coverage can be computed from the recommender’s outputs alone, diversity from item features such as composition, and ranking quality from independently collected expert relevance judgements rather than from purchase logs. What is unavailable is evaluation against observed local purchasing preference, which is one form of evidence rather than the whole of it. Chapter 8 therefore reports coverage, constraint-violation rate, unsupported-claim rate and abstention rate alongside expert judgement.

## 2.3 The skincare domain

### 2.3.1 Ingredient nomenclature

Cosmetic ingredients are labelled using the International Nomenclature of Cosmetic Ingredients, a standardised naming system adopted for ingredient labelling in the European Union and several other jurisdictions. Two properties of an INCI list are used directly in this thesis. It is *standardised*, so the same substance carries the same name across manufacturers, which makes matching against a register feasible. And above one per cent it is *ordered by descending concentration*, so the position of an ingredient carries information that an unordered set representation would discard. The similarity measure in Chapter 7 uses that ordering.

### 2.3.2 Skin type classification

Consumer skincare uses several classification schemes. The Fitzpatrick phototype describes response to ultraviolet exposure and is not a guide to product selection. The scheme relevant here is the Baumann Skin Type Indicator, which classifies skin along four independent dichotomies: dry or oily, sensitive or resistant, pigmented or non-pigmented, and wrinkle-prone or tight [19]. The instrument has been validated across a large population and in several languages [20], and is one of the few self-report skin classification instruments for which validation data is published [78].

The relevance is direct rather than decorative. The skin representation used in this thesis carries a dry/oily axis and an independent sensitive/resistant axis, which are the first two Baumann dichotomies. Aligning to a validated clinical instrument, rather than to a vocabulary assembled from retailer copy, is what allows the skin representation to be checked by someone outside this project.

[note: This alignment is worth stating explicitly in Chapter 5 as well. It partially answers the limitation that the concern vocabulary has not been reviewed by a clinician: the skin-type axes are taken from an instrument that clinicians already validated.]

## 2.4 The regulatory setting

### 2.4.1 Regulation (EC) No 1223/2009

Cosmetic products placed on the European market are governed by Regulation (EC) No 1223/2009 [40], which operates through a set of annexes. Annex II lists

substances prohibited in cosmetic products. Annex III lists substances restricted, with the conditions and maximum concentrations under which each may be used. Annexes IV, V and VI list the permitted colourants, preservatives and ultraviolet filters respectively. A restriction is therefore not a general judgement about an ingredient but a specific, citable entry.

The Commission maintains CosIng, the cosmetic ingredient database, which records for each substance its INCI name, its CAS number, the function the Commission recognises, and its regulatory status under the annexes. CosIng is the authority to which the ingredient layer of this work is linked, and it is the reason a statement such as *this product contains a restricted preservative* can be traced to a legal instrument rather than to a rule invented for this thesis.

### 2.4.2 Fragrance allergens and the 2023 amendment

A subset of fragrance substances must be declared individually in the ingredient list rather than being covered by the collective term *parfum*, above 0.001 % in leave-on products and 0.01 % in rinse-off products. This list has historically comprised 26 entries: 24 individually named chemical substances together with two natural extracts, *Evernia prunastri* and *Evernia furfuracea*.

Commission Regulation (EU) 2023/1545 [39] substantially expands this list, adding substances identified by the Scientific Committee on Consumer Safety and bringing the total requiring individual declaration to approximately 80. Products newly placed on the market must comply from 31 July 2026, and products already on the market from 31 July 2028.

*A moving target, and what it means for this work.* The allergen list changed during the period of this research, and the first compliance deadline has now passed. The dataset described in Chapter 4 was constructed against the historical list of 26. Two consequences follow and are handled explicitly rather than quietly: the allergen analysis reports which list it was computed against, and the ontology models the allergen set as a *versioned* authority-issued list rather than as a fixed class, so that an update to the register is a data change and not a model change.

**[TODO: Verify the exact counts against the consolidated text of Annex III before submission, and state in Chapter 4 which list version the allergen columns were computed against. The 24-versus-26**

discrepancy is a counting convention, not a disagreement: state your convention once.]

## 2.5 The Lebanese market

Three features of the Lebanese setting bear directly on the design of the system, and all three are documented in the health-systems literature rather than assumed.

**Import dependence.** Lebanon imports approximately 95% of its pharmaceutical products [58]. Dermocosmetic products are sold through the same pharmacy channel and subject to the same import conditions, so local availability is contingent on a supply chain that has repeatedly been interrupted.

**Currency instability.** The Lebanese pound lost the greater part of its value from 2019 onwards [91], and subsidy arrangements were subsequently withdrawn [37, 58]. A price is therefore meaningful only in conjunction with a currency and a date, which is why price is modelled in this work as a property of an offer rather than of a product.

**Availability as a search problem.** Shortages are sufficiently routine that locating a product has become a distinct task from choosing one, and digital platforms have been proposed specifically to tell a user which pharmacy currently holds a given item [17]. This is the clearest evidence that availability belongs in the model: in this market it is already treated as information worth building a system around.

These conditions do not change what a good skincare recommendation is. They change what a usable one is, and the remainder of the thesis is concerned with the difference.



## Chapter 3

# Related Work

### 3.1 Scope and method of the review

This chapter reviews the work against which the contribution is measured. It is ordered as the work was carried out. Section 3.2 reviews how skincare product datasets and domain resources are constructed, because the dataset of Chapter 4 was built before anything else and the decision of what to reuse was taken against those resources. Section 3.3 reviews skincare recommender systems, the direct competitors. Section 3.4 reviews knowledge-based recommendation in other domains, the source of the mechanisms used here. Section 3.5 reviews ontology engineering. Sections 3.6 to 3.9 follow the remaining stages of the methodology and Section 3.10 states the gap.

Literature was identified in two passes: general search over the terms of the domain, then an academic index covering Semantic Scholar, PubMed, Scopus and arXiv, which returned nine works the first pass missed.

Two inclusion rules apply, differing by the role a work plays. A work cited to justify a methodological decision is drawn from a first-quartile journal, an A or A\* conference, or is a canonical reference. A work that performs skincare recommendation is included whatever its venue, since a competing system cannot be omitted because its journal is weak; where the venue is weak this is stated at the point of citation. Work published before 2015 is cited only where it originates a concept still in use.

### 3.2 Skincare datasets and domain knowledge resources

Every system reviewed in Section 3.3 rests on a catalogue, and in most cases the catalogue is the least documented part of the work. This section reviews the

resources that exist independently of any recommender, since these are what the dataset of Chapter 4 was built against.

### 3.2.1 Regulatory and toxicological resources

CosIng is the European Commission’s inventory of cosmetic ingredients kept under Regulation (EC) No 1223/2009 [38, 40]. It records for each substance an INCI name, CAS and EC numbers, the functions the Commission recognises, and status under the annexes. It is distributed as a web application and a table, not as a graph.

CosIng-KG republishes the register as RDF with resolvable identifiers per substance [23]. This work links to it. TOXIN, in *Database*, assembles toxicological evidence for cosmetic ingredients for animal-free risk assessment [89]. CCIBP, in *Bioinformatics*, adds structural and bioactivity data [49]. EUCOSMETICS, distributed through the NORMAN Suspect List Exchange, provides a combined ingredient inventory for analytical screening [101]. All four are rigorous about substance identity and none contains a product a consumer can buy.

Two results determine how allergen status is modelled here. Białas et al. [22] survey the regulatory status of skin sensitisation and describe the mechanism by which the declarable fragrance allergen list is revised, which produced the expansion in Commission Regulation (EU) 2023/1545 [39]. Klaschka [61] shows that substances of natural origin are over-represented among declarable allergens. The allergen set is therefore modelled as a versioned, authority-issued list, and no inference is drawn from a natural-origin claim.

### 3.2.2 Clinical and consumer vocabularies

DermO is an ontology of dermatologic disease in the *Journal of Biomedical Semantics*, distributed through BioPortal in OBO and OWL 2 with lexical variants for clinical terms [1]. Its granularity is diagnostic, so only its upper region is usable by a retail recommender, where consumer concerns such as acne have clinical counterparts that can be referenced without importing the hierarchy beneath them. D3X links dermoscopic patterns to differential diagnoses [4] and was rejected: a resource organised around what a clinician sees through a dermatoscope has no point of contact with a product catalogue.

The Baumann Skin Type Indicator classifies skin along four independent dichotomies and is validated across a large population and several languages [19, 20, 78]. Two of the four, dry–oily and sensitive–resistant, are the axes this dataset

already carries as separate fields, so the skin representation aligns to a published instrument rather than to retailer vocabulary.

### 3.2.3 Cosmetics ontologies outside recommendation

OntoCosmetic is the most developed cosmetics ontology in the literature, built by Serna et al. [90] for decision support in emulsion formulation and extended into working software by Gabriel et al. [45]. It models phases, functions and process constraints, and is published under a persistent identifier. Its user is a formulator, not a consumer, which is why it is reviewed here rather than beside the recommenders: it supplies vocabulary for formulation properties and contains no marketed products, prices or consumer-stated concerns.

The flavouring ontology of hal [5] derives halal status for a compound from the status of its constituents. The inferential shape is the one used in Chapter 5 to derive regulatory status for a product from the status of its ingredients, under a different authority.

### 3.2.4 Construction practice and what it requires

Two methodological literatures govern how the dataset was built. Record linkage supplies the procedure for deciding that a local listing and an international record describe one product: blocking to make the comparison tractable [81], string similarity for the comparison itself [30], and a two-threshold decision rule with a review region between acceptance and rejection [41]. That procedure is applied in Section 4.4.

Publication practice supplies the rest. A resource intended for reuse carries machine-readable provenance for each claim [64], expresses its controlled lists as concepts rather than strings [73], and describes itself in a standard catalogue vocabulary [36, 102]. None of the skincare resources above meets all three, and none records which agent asserted a given suitability claim.

### 3.2.5 What is reused and what is built

Every resource with regulatory grounding contains no products. Every system in Table 3.2 that contains products has no regulatory grounding. Neither side is deficient on its own terms: a substance register has no reason to carry retail prices, and a national recommender has no obvious route into a Commission database. The consequence is that no existing resource can answer whether a product a user

**Table 3.1.** Knowledge resources reviewed, the capability each supplies, and the decision taken. *Reuse* imports or aligns to the vocabulary; *reference* follows MIREOT, citing individual terms without importing the source; *cite* acknowledges a pattern without adopting the artefact.

| Resource           | Kind           | Capability supplied                                                                                               | Decision  | Products       |
|--------------------|----------------|-------------------------------------------------------------------------------------------------------------------|-----------|----------------|
| CosIng [38]        | Register       | Ingredient identity, function, regulatory status                                                                  | Reuse     | —              |
| CosIng-KG [23]     | RDF export     | The register as a graph                                                                                           | Assess    | —              |
| TOXIN [89]         | KG             | Construction pattern; named graphs per source                                                                     | Cite      | —              |
| CCIBP [49]         | Platform       | Non-EU regulatory status                                                                                          | Reference | —              |
| DermO [1]          | Ontology       | Skin concerns with clinical standing                                                                              | Reference | —              |
| BSTI [19]          | Instrument     | Validated skin-type axes                                                                                          | Align     | —              |
| OntoCos-metic [90] | Ontology       | Ingredient function vocabulary; formulation modelling                                                             | Align     | — <sup>b</sup> |
| <b>This work</b>   | <b>Dataset</b> | <b>Ingredient identity and regulatory status reused from CosIng; skin axes aligned to BSTI; anatomy to UBERON</b> | <b>—</b>  | <b>12,629</b>  |

The final column is the point of the table. Every resource that supplies regulatory or clinical grounding contains no products; every system that contains products (Table 3.2) supplies no grounding.

<sup>b</sup> The published OWL file declares 283 named individuals of heterogeneous types, including ingredients, heuristics and five formulation instances. None is a marketed retail product, so no product count is given.

Selected-term reuse follows MIREOT: named terms are referenced with their provenance rather than the source ontology being imported whole. Importing, aligning, referencing an IRI and citing a modelling pattern are distinct decisions, and the decision column names which was taken.

can buy this week contains a substance the law restricts, and none records who asserted that the product suits that user.

### 3.3 Skincare recommendation and related prediction tasks

The studies below differ in what they take as input, what they predict, and what their evaluation establishes. Some recommend a product to a user, some predict an attribute of a product, and the two require different evidence. They are separated here on that basis rather than by technique.

#### 3.3.1 Knowledge-based and hybrid recommenders

Moe and Aung [74] treat product selection as a *cross-domain* problem. Two ontologies are built, one of facial skin problems and one of cosmetics, and the recommendation is produced by bridging concepts between them. Retrieval within each domain uses taxonomic conversational case-based reasoning, and the cross-domain link is computed with the Ford–Fulkerson maximum-flow algorithm over a graph connecting a source concept to a target concept. The cosmetics ontology carries `hasIngredients` and `hasIngredientsValue` data properties, so ingredients are represented, and evaluation reports precision, recall and F-measure as a function of a weighting constant. What the model does not carry is any external referent for those ingredient values: they are strings local to the ontology, with no link to a register and no regulatory status.

Hansanie and Silva [54] combine a convolutional network with an ontology. The network grades acne severity from a facial photograph, and a Protégé ontology relates user profile, skincare information and product information; recommendation is driven by semantic similarity over those concepts, taking skin type, concerns, acne severity and allergy ingredients as input. The system was evaluated by 24 participants and reports 87.5% accuracy. The study is the closest precedent in this review for combining a learned perceptual component with symbolic recommendation. Its evaluation is a small user study rather than a held-out benchmark, and the participant count bounds what the accuracy figure supports.

Abesová et al. [8] is a course report from Vrije Universiteit Amsterdam rather than a peer-reviewed publication, and is included because of what it implements. The user supplies skin type, skin tone, residential country and a desired routine complexity, and the system returns a personalised routine together with the local Sephora web link and nearby physical stores, derived from a mapping of countries in which Sephora operates. Ingredient-level restrictions are identified in its

own limitations as future work, so the system does not score products against ingredient sensitivities.

*A precedent this thesis must credit.* Abesová et al. [8] represents geographic availability: it knows which countries a retailer serves and returns a local link. Any claim in this thesis about availability must therefore be stated as a narrower and more specific one, namely seller-level listing of an individual product with an observation date, and that claim carries its own evidential burden, examined in Section 4.9.2.

Jannatin and Ratnasari [59] is the closest published system to the one proposed here and is reported from the full text. The ontology is built under METHONTOLOGY, named explicitly, inside a four-stage development cycle; requirements were derived from a literature review and a user survey and then checked against dermatological consultation and product-labelling guidance. It comprises twelve top-level classes — among them `Formulation Trait` and `What It Does`, a distinction between a physical property of a formulation and a claim made about it — with more than twenty-five object properties and eighteen data properties. The knowledge base was populated by scraping two Indonesian retail platforms for products and a third source, SkinSort, for ingredient records, giving more than 3,800 product entries and more than 28,000 ingredient records; the latter figure counts scraped records, not distinct substances. Population was by script, with integration through the OWL API and no declarative mapping language. Inference is performed by Apache Jena Fuseki over user input converted to RDF, traversing `suitableForSkinType`, `treatsConcern`, `incompatibleWith` and `avoidFor`. The task is therefore posed as constrained selection rather than as ranking, which is the same formulation adopted in Chapter 7.

Three properties of that system bear directly on the claims made in this thesis. First, it records a product-level regulatory identifier, the Indonesian BPOM registration number, as a data property. A claim that no reviewed recommender carries regulatory information is therefore not tenable, and the defensible claim is narrower: a registration identifier attests that a product is registered, whereas the linkage required here is from each ingredient mention to a statutory register entry carrying its recognised function and any annex restriction. Second, ingredient interactions are modelled, through `synergisticWith`, `incompatibleWith`, `mayAggravateConcern` and an `InteractionWarnings` data property, so the system is not confined to one-to-one mappings; what the publication does not describe is any record of which authority asserts a given interaction, so the graph cannot

distinguish an asserted incompatibility from a sourced one. Third, evaluation is confined to the Technology Acceptance Model with ten respondents, reporting a mean of 4.5 of 5 across perceived usefulness and perceived ease of use, with no accuracy measure, no baseline and no reference set; the discussion nevertheless asserts superiority over content-based and collaborative filtering, which the reported measurements do not support. The ontology is not deposited in any repository.

*A counted finding, not an impression.* Across the skincare recommenders reviewed in this section, not one reports recommendation accuracy against a reference set. Jannatin and Ratnasari [59] measures acceptance with ten respondents and Hansanie and Silva [54] satisfaction with twenty-four; Abesová et al. [8] presents a single worked example; Moe and Aung [74] plots precision and recall without stating a dataset or a baseline. This matches the finding of Rahayu et al. [86] that none of twenty-eight primary studies described an ontology evaluation methodology, and it is the reason the evaluation design in Chapter 8 is derived from a published evaluation framework rather than from a user-satisfaction instrument.

Utari et al. [99] remains recorded as requiring an original-source audit before its method and counts are reported. It has an identifiable bibliographic record and is retained in the comparison with its cells coded as unaudited. That record describes ontology construction with a stated number of individuals, which is not the same quantity as a count of marketed products, and the two must not be conflated.

### 3.3.2 Learned ingredient and skin models

Lee et al. [65] encode the ingredient list as an *ordered* sequence with a Transformer and combine it with facial skin analysis, over a catalogue of roughly 8,000 products and approximately 11,000 distinct ingredient names. The paper discusses ingredient order and sensitivity-related skin characteristics explicitly. Two consequences follow for this thesis. Order-aware ingredient representation and a sensitivity axis are established in the literature and cannot be presented here as first uses; what remains open is whether the ingredient identifiers are grounded in an external register and whether the resulting claims carry a source.

Liu et al. [67] address ingredient-based attribute extraction with an explanation objective, using an energy-based implicit architecture over 11,580 data points,

released as a preprint. The task is supervised extraction of product attributes from composition rather than ranking, which makes it a close precedent for the enrichment stage of this thesis rather than a competitor at the recommendation stage.

Hoang et al. [57] introduce HaCKG, a knowledge graph with relation-aware pre-training whose evaluated downstream task is predicting the halal status of daily products. The graph represents allergy and restriction-state properties. It is compared here as a related prediction task: the inferential shape, deriving a product-level status from the status of its constituents under a named authority, is the same shape this thesis applies to European cosmetic regulation, and whether its restriction states are linked to a statutory identifier is a separate question requiring verification.

### 3.3.3 Comparative synthesis

Table 3.2 records what each study set out to do, with what data and against what evaluation. Table 3.3 records capabilities, using codes that keep four states apart: a capability demonstrated in the paper, one the paper establishes is absent, one the paper does not report, and one awaiting an original-source check. Coding a silence as an absence would manufacture a gap, so silence is coded as silence. One study carries an identifiable bibliographic record whose full text has not been audited, Utari et al. [99]; its cells are coded **U** throughout rather than removed, because an inaccessible publisher page is not evidence that a study does not exist. Jannatin and Ratnasari [59] was coded **U** in an earlier draft and is now coded from the published article. Utari’s published count is of ontology individuals rather than marketed products, so no product figure is carried into Table 3.2. The supporting passage and page for every coded cell are recorded in Appendix B.

Read across the studies, three requirements are addressed only in part. Ingredient representation is present in most of them. A product-level regulatory registration identifier is recorded by Jannatin and Ratnasari [59], but a link from an individual ingredient mention to a statutory register entry, with the recognised function and any annex restriction attached, is not documented in any of the reviewed publications. Availability is represented at country level by Abesová et al. [8], and seller-level listing is not documented in any of them. Nor does any of the publications describe recording which agent asserted a given suitability claim, which is what would separate a manufacturer’s statement from an inference made by the system’s authors.

**Table 3.2.** Reviewed studies by task, data and evaluation. Studies that recommend a product to a user and studies that predict a product attribute are separated, because they require different evidence. Counts are as each paper reports them; where a paper reports a count of ontology individuals rather than marketed products, no product figure is given.

| Study                                | Task                         | Method                                                                   | Data reported                                     | Evaluation                            |
|--------------------------------------|------------------------------|--------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------|
| <i>Recommend a product to a user</i> |                              |                                                                          |                                                   |                                       |
| Moe and Aung [74]                    | Cross-domain recommendation  | Two ontologies; taxonomic conversational CBR; Ford–Fulkerson bridge      | Not reported                                      | Precision, recall, F-measure          |
| Hansanie and Silva [54]              | Product recommendation       | CNN acne grading + Protégé ontology; semantic similarity                 | Not reported                                      | 24-participant study, 87.5 %          |
| Abesová et al. [8]                   | Routine + local retailer     | Ontology; country-to-retailer mapping                                    | Sephora review export                             | Course report; no measured evaluation |
| Utari et al. [99]                    | Case retrieval               | Ontology construction                                                    | Individuals reported, not products                | Not audited                           |
| Jannatin and Ratnasari [59]          | Product recommendation       | METHONTOLOGY ontology, 12 classes; Jena Fuseki inference over user input | 3,800+ products; 28,000+ ingredient records       | TAM, 10 participants, mean 4.5/5      |
| <i>Predict a product attribute</i>   |                              |                                                                          |                                                   |                                       |
| Lee et al. [65]                      | Ingredient-based suitability | Transformer over <i>ordered</i> INCI + facial analysis                   | ≈8,000 products; ≈11,000 ingredients              | <b>[TODO: task, metric, result]</b>   |
| Liu et al. [67]                      | Attribute extraction         | Energy-based implicit model; explanation objective                       | 11,580 data points                                | Extraction metrics (preprint)         |
| Hoang et al. [57]                    | Halal status prediction      | Relation-aware KG pre-training                                           | Not reported                                      | Link-prediction metrics               |
| <b>This work</b>                     | Dataset; planned recommender | See Chapter 4; Chapter 7 planned                                         | 12,629 products, 1,463 brands, 14,520 ingredients | Chapter 8, planned                    |

A university course report, not a peer-reviewed publication. Included because of the capability it implements.

**Table 3.3.** Capabilities of the reviewed systems. **P** demonstrated in the paper; **o** present in part, with the extent given in a note; **X** the paper establishes the capability is absent; **–** not documented in the publication; **U** unaudited, awaiting an original-source check. The final two columns separate what this thesis has delivered in the dataset from what Chapters 5 to 8 plan. Supporting passage and page for every coded cell are recorded in Appendix B.

| Capability                      | Moe | Han. | Abe. | Uta. | Jan. | Lee | Liu | HaC. | Ours<br>del. | Ours<br>pl. |
|---------------------------------|-----|------|------|------|------|-----|-----|------|--------------|-------------|
| <i>Formulation</i>              |     |      |      |      |      |     |     |      |              |             |
| Ingredients represented         | P   | P    | X    | U    | P    | P   | P   | P    | P            | —           |
| Ingredient <i>order</i> used    | –   | –    | –    | U    | –    | P   | –   | –    | P            | —           |
| Linked to a regulatory register | –   | –    | –    | U    | o    | –   | –   | –    | P            | —           |
| Restriction status represented  | –   | –    | –    | U    | o    | –   | –   | P    | P            | —           |
| <i>Suitability</i>              |     |      |      |      |      |     |     |      |              |             |
| Skin type modelled              | P   | P    | P    | U    | P    | P   | –   | –    | P            | —           |
| Sensitivity as a separate axis  | –   | –    | –    | U    | P    | P   | –   | –    | P            | —           |
| <i>Provenance</i>               |     |      |      |      |      |     |     |      |              |             |
| Source recorded per claim       | –   | –    | –    | U    | –    | –   | –   | –    | P            | —           |
| Evidence strength graded        | –   | –    | –    | U    | –    | –   | –   | –    | P            | —           |
| <i>Market</i>                   |     |      |      |      |      |     |     |      |              |             |
| Country-level availability      | –   | –    | P    | U    | –    | –   | –   | –    | P            | —           |
| Seller-level product listing    | –   | –    | –    | U    | –    | –   | –   | –    | P            | —           |
| Price recorded                  | –   | –    | –    | U    | –    | –   | –   | –    | P            | —           |
| Price observation date          | –   | –    | –    | U    | –    | –   | –   | –    | o            | P           |
| <i>Output</i>                   |     |      |      |      |      |     |     |      |              |             |
| Reasoner applied                | –   | P    | P    | U    | P    | –   | –   | –    | –            | P           |
| Explanation produced            | –   | P    | P    | U    | –    | –   | P   | –    | –            | P           |
| Alternatives retrieved          | –   | –    | –    | U    | –    | –   | –   | –    | –            | P           |

Counted as present only where a sensitivity dimension is modelled independently of the dry–oily dimension, as in the first two axes of the Baumann instrument [19]. Coded **o** (partial) for the delivered dataset rather than absent: an observation date is present for 5,453 catalogue prices and for 16 of the local records (Section 4.9.2). The adequacy requirement this thesis sets is a date on every *local* price, and the release does not meet it.

Jannatin and Ratnasari [59] is coded **o**: a product-level registration identifier (Indonesian BPOM number) is recorded as a data property, but no ingredient mention is linked to a statutory register entry.

Jannatin and Ratnasari [59] is coded **o**: allergen types and ingredient interactions are represented, but the categories are defined by the authors rather than drawn from a statutory annex, and no authority is recorded against any asserted interaction.

These are statements about what the publications document, not proofs of absence. A system may implement a capability its paper does not report, and the codes in Table 3.3 keep that distinction: – marks silence in the source, and **X** is used only where a paper states that a capability is out of scope.

The final two columns of Table 3.3 are deliberately separate. What this thesis has delivered is the dataset layer; the reasoner, the explanations and the retrieval of alternatives are designs in Chapters 5 to 7 rather than results. One delivered cell is coded as an absence rather than a presence: the price observation date, which the argument of this thesis requires and the current release does not supply for local prices.

## 3.4 Knowledge-based recommendation in other domains

The mechanisms this work uses were developed outside skincare, in domains facing the same conditions: a catalogue whose items carry consequences, sparse or absent interaction data, and a requirement to state a reason.

### 3.4.1 What a knowledge graph provides

Guo et al. [52], in *IEEE TKDE*, divide knowledge-graph-based recommendation into embedding-based, path-based and unified methods, and state the trade-off this work accepts: semantic methods forgo the accuracy abundant interaction data would give, and gain the ability to operate with none. Zhang et al. [110] reach the same division from the deep-learning side.

CKE [109], RippleNet [104] and KGAT [105] incorporate graph structure into a learned representation with increasing sophistication, and all require a user-item interaction matrix. KPRN [106] reasons over explicit paths and returns the path as justification. The path formulation is the relevant one here, since a path is inspectable and an embedding is not.

Balloccu et al. [18] show that post-hoc explanations over a knowledge graph can be faithful to the model or plausible to the reader without being both, and Spillo et al. [94] report the same tension for neural recommenders. Shokrzadeh et al. [93] generate the recommendation and its path jointly. The structural lesson is that an explanation produced by a different mechanism from the rec-

ommendation describes the output rather than justifying it, which is why the recommendation in Chapter 7 is the reasoner’s own derivation.

### 3.4.2 Domain systems and the mechanism each contributes

*Food.* FoodKG builds a knowledge graph of recipes, ingredients and nutrition supporting question answering and constrained recommendation [55]. The domain shape transfers directly: an item is a composition of substances and a constraint applies at substance level. Chen et al. [29] show that a dietary restriction must be a hard filter rather than a weighted feature, since a learned penalty can be outweighed. The equivalent statement here is that a declared allergen is not a preference, implemented as a defined class in Chapter 5.

*E-commerce.* AliCoCo connects the language shoppers use to the catalogue they shop in [68], establishing consumer-facing concepts as a layer distinct from both product taxonomy and clinical vocabulary. A user asking for something for dull skin is naming neither a category nor a diagnosis.

*Retail ontologies.* Alaa et al. [11] evolve the ontology as the catalogue changes. Tiryaki et al. [98] bind an ontology-based recommender to live e-commerce data, which establishes availability as data flowing into the model rather than an assumption made outside it. Availability is held as a current state with no record of when it was observed, which is insufficient in a market where a price is valid for a fortnight.

*Tourism and e-learning.* Middleton et al. [72] originated ontological user profiling, a profile expressed in domain terms and therefore inspectable. Tarus et al. [96, 97] report that the semantic component contributes most where behavioural data is thinnest, the condition assumed throughout this thesis. Lahoud et al. [63] review e-learning recommendation under regional deployment constraints.

*Linked data.* Di Noia et al. [33] and Ostuni et al. [79] established recommendation over publicly linked data using external identifiers to describe items, which licenses the approach taken here: an ingredient is identified by its register entry, and the properties reasoned over are the register’s.

The mechanisms assembled are a semantic item representation that works without interaction data, hard constraints expressed as logic, an explicit consumer-language layer, availability as dated data, and explanations produced by the mechanism that produces the recommendation. No system in Section 3.3 assembles more than two. What this literature does not supply is the regulatory

dimension, since no dietary authority publishes a register in the form the cosmetic regulator does.

## 3.5 Ontology engineering for recommendation

Stage 1 builds an ontology. This section supplies the construction method, the reuse practice, and the evaluation criteria. It follows Section 3.2 because the ontology here was designed over data that already existed, which is a different starting condition from the one most of these methodologies assume.

### 3.5.1 Construction methodologies

Grüninger and Fox [51] introduced competency questions: the ontology is specified by the questions it must answer, so a question that cannot be answered locates a defect precisely. None of the systems in Section 3.3 reports any. Keet and Alharbi [60] show that many published competency questions cannot be turned into a query at all, and Quirino et al. [85] give patterns that determine the mapping to a query in advance. Both are applied in Chapter 5, where each competency question is paired with the SPARQL query answering it.

METHONTOLOGY established the development lifecycle [42] and Noy and McGuinness [76] remains the most cited practical guide; its second step addresses reuse of existing ontologies explicitly, so neither treats construction from nothing as the only path. They are not adopted here for a different reason: neither prescribes the artefacts this project needs to produce, namely a versioned published vocabulary with per-term provenance for imported terms, and neither structures development around modules that can be developed and tested separately.

Two current methodologies do. Linked Open Terms structures development as iterations over requirement specification, implementation, publication and maintenance, treating publication as a stage of the method [84]. Modular Ontology Modeling builds the ontology as modules assembled from patterns, each developed against its own competency questions [92], drawing on the pattern literature of Blomqvist et al. [25] and Hitzler et al. [56]. They are complementary: LOT governs process, MOMo governs internal structure. O’Sullivan et al. [80] confirm that the module boundary is what keeps a large ontology maintainable, and Braun et al. [26] report the same where the ontology evolves alongside its dataset.

### 3.5.2 Reuse and modularity

Grau et al. [50] give the formal account of importing part of an ontology without importing unintended consequences. Halper et al. [53] and Ochs et al. [77] report empirically that wholesale import of a large vocabulary is common in biomedical ontologies and that it degrades both reasoning performance and comprehensibility. The alternative they describe, importing the minimum set of terms with recorded provenance, is what Chapter 5 does for the terms taken from DermO.

UBERON supplies cross-species anatomy with stable identifiers [75] and is the target for the anatomical field of Section 4.8.5. ChEBI supplies chemical entities [69] and was set aside for the ingredient layer, since a cosmetic label asserts an INCI name rather than a chemical structure; it returns where structural similarity is needed. FoodOn shows that a consumer-facing domain can be modelled to clinical standards [34], and is the closest precedent for bridging register nomenclature and shopper description inside one vocabulary.

PROV-O supplies the terms for the evidence model [64], SKOS for the concern vocabulary with multilingual labels [73], and DCAT and Dublin Core for the artefact's self-description [36, 102]. Expressing a local evidence scale in PROV-O rather than in invented properties is what allows other provenance tooling to read it.

### 3.5.3 Evaluation practice

Construction and evaluation fail together: an ontology built without competency questions has nothing to be evaluated functionally against.

OOPS! detects modelling pitfalls [83] and FOOPS! scores an ontology against the FAIR principles [47]. Both are structural and neither measures whether the domain is modelled correctly, which is what executing the competency questions answers.

Rahayu et al. [86] find that the ontology-based recommender literature reports user satisfaction far more often than whether the system does what it was specified to do. Zangerle and Bauer [107] give the framework adopted here, separating what is evaluated from how and against what. Their central observation applies to every system in Section 3.3: an accuracy figure computed against held-out interactions from one catalogue says nothing about behaviour in a market where those interactions do not exist. Where no interaction data is available the question is whether the system answers its specified questions, produces derivations a domain expert accepts, and remains consistent under reasoning.

## 3.6 Knowledge graph population

[TODO:Written during Stage 2. Reviews declarative mapping from tabular sources to RDF: R2RML as the W3C recommendation [31], RML and its engines for non-relational input, and Morph-KGC [15] for mapping-partition scaling. Reviews validation of the populated graph with SHACL [62] and the separation of asserted from inferred statements into named graphs. Closes on what the reviewed work leaves open for a graph whose source records carry unequal evidence.]

## 3.7 Neural and language-model enrichment

[TODO:Written during Stage 3. Reviews graph representation learning (RDF2Vec [88], OWL2Vec\* [28]), language models for ontology learning and term extraction (LLMs4OL [16], Li et al. [66]), and retrieval-augmented generation over graphs [12]. The section must state the validation condition used here: a term proposed by a model enters the graph only after it resolves against an authoritative source, and its evidence level records that it was machine-proposed.]

## 3.8 Semantic reasoning for recommendation

[TODO:Written during Stage 4. Reviews reasoning-based recommendation from Blanco-Fernández et al. [24], García-Crespo et al. [46] and Picot-Clément et al. [82] through to current constraint-aware work, and the use of defined classes and SWRL rules to derive suitability. Closes on the treatment of hard constraints against ranked preferences, following Chen et al. [29].]

## 3.9 Ingredient similarity and substitution

[TODO:Written during Stage 5. Reviews set-based and order-aware similarity over composition lists, chemical similarity via ChEBI [69], and substitution as a recommendation task in food and pharmacy. The distinguishing requirement here is that INCI order carries concentra-

tion information above one per cent, so an unordered set measure discards evidence that is present.]

### 3.10 Summary and the gap addressed

[TODO: Written once Sections 3.6 to 3.9 are complete. Consolidates the closing paragraphs of each section into a single statement of the gap, and presents the master matrix combining Table 3.2 and Table 3.1 along the dimensions that separate this work: regulatory grounding at ingredient level, recorded provenance for every suitability claim, availability and dated price as modelled properties, and substitution of an unobtainable recommendation.]

Part II

# Contributions



## Chapter 4

# The Skincare Dataset

### 4.1 Purpose and scope

Section 3.2 established that no existing resource carries both marketed products and regulatory grounding. This chapter reports the resource built to close that gap: 12,629 skincare products from 1,463 brands across 22 product types, described by 111 fields, of which 5,014 are listed by Lebanese retailers and 802 are made by Lebanese manufacturers.

The dataset is reported before the ontology for a reason that shaped both. It was constructed first, so the ontology of Chapter 5 models data that exists rather than a domain in the abstract, and every class and property in it is populated. The competency questions of Chapter 8 are answerable because the fields they interrogate were built before the questions were written.

Three commitments govern the construction and recur throughout the chapter. Every value records where it came from. Nothing is asserted about the absence of an ingredient, only about its presence. And a technical failure is never written into the data as a finding about a product.

### 4.2 Sources

The dataset merges three source categories, kept separate in the `source_category` field rather than flattened, because a question about the Lebanese market and a question about the international catalogue have different denominators.

#### 4.2.1 The international catalogue

**[TODO: Write this subsection to the same standard as the two below it: the source name and version, the collection period, the original size before filtering, the retained size, the fields inherited from it, and the separate review-matching pass that attached ratings and review**

**Table 4.1.** The three source categories and their sizes. *Both* marks a product present in the international catalogue and also listed by a Lebanese retailer, which is the segment on which international product knowledge and local availability can be compared directly.

| Category        | Products      | What it is                                                    |
|-----------------|---------------|---------------------------------------------------------------|
| Global          | 6,294         | International catalogue, matched to reviews and ratings       |
| Lebanese retail | 5,014         | Listings from Lebanese online pharmacies and beauty retailers |
| Lebanese origin | 802           | Products made by Lebanese manufacturers                       |
| Both            | 519           | Present in the international catalogue and listed locally     |
| <b>Total</b>    | <b>12,629</b> |                                                               |

**text to catalogue products. The international catalogue supplies 6,294 products and is currently the least documented of the three streams, which is the largest remaining reproducibility gap in this chapter.]**

### 4.2.2 The Lebanese retail source

Lebanese retail listings were collected from nine online shops, of which five account for almost all of the catalogue: Sohati Care, Feel22, Mazen Online, Zeina Care and Nexus Care. The same product frequently appears in more than one shop, which is a property of the market rather than a defect in the data, and it is preserved: `shops_in_lebanon` records how many shops list a product, and 356 products are listed by more than one. A product sold by four shops and a product sold by one are different objects from the point of view of availability, and collapsing them would destroy the only signal this dataset has about supply.

### 4.2.3 The Lebanese origin source

Products made in Lebanon are not listed as such anywhere, so the brands had to be found before the products could be collected. Searching was carried out in three languages and sideways as well as directly, since a Lebanese soap maker is more often named in a craft directory than in a beauty listing.

Collection reached 65 sites and harvested 38,835 candidate products, of which 1,647 were retained. A rejection rate of this size has to be defensible one product at a time, so every rejected row carries a written reason and is kept in a file rather than deleted. Three tests were applied.

1. **Is this a brand or a shop?** A site where one brand name accounts for at least sixty per cent of the catalogue is a brand. A site carrying two hundred and forty-seven brands is a retailer, and its products belong to the retail set.
2. **Is this skincare?** Lebanese brands sell soap beside candles, olive oil and food. Only products falling in the categories kept for the international catalogue are retained.
3. **Is the brand Lebanese?** A +961 telephone number, a .lb domain, a Lebanese address, or Lebanon named on the page.

The three tests were applied at different stages and their counts do not partition a single population: 27 sites passed all three, 40 were rejected outright, and a further twenty-five were split product by product, so a site may be counted in more than one of those figures. Of the sites rejected outright, nineteen failed the first test as shops rather than brands.

**[TODO: Add a collection-flow table with stage-specific denominators showing how 65 sites and 38,835 candidate products became 1,647 retained records and then 802 Lebanese-origin rows after deduplication and merge. The two retained figures differ and the chapter must say where the 845 records went.]** Two rejections are worth stating because they show why the tests are needed: two sites carry a Lebanese domain name and pass every test about being in Lebanon, yet one sells a British brand and the other a Turkish one. A domain name is not evidence about who makes a product. A further twenty-five sites were split rather than accepted or rejected whole, because a genuine Lebanese skincare brand may also sell things that are not skincare, and those catalogues were judged product by product.

*Why the exclusions are published.* The retained set is four per cent of what was collected. A reader who cannot see the ninety-six per cent has to take the four on trust. The excluded rows are therefore published with their reasons alongside the dataset, in files separating non-skincare products, makeup, multi-product sets, bundles and non-products.

## 4.3 Cleaning

Cleaning was not a single pass. Each fault in Table 4.2 was found after the dataset appeared finished, and each had produced a plausible value rather than an error, which is why no check written earlier caught it. The order of operations follows

the standard rule for multi-source integration: each source is cleaned completely before it is joined to anything else, so that a failure to match cannot be caused by a difference the cleaning would have removed [87].

**Table 4.2.** Faults found after the dataset first appeared complete. Each produced a plausible value rather than an error, so none was caught by a check written earlier. The two recurring classes are rules that matched text without reading it, and labels that meant more than one thing.

| Fault                             | What it produced                                                | Scale                          |
|-----------------------------------|-----------------------------------------------------------------|--------------------------------|
| Negation not tested               | <i>“not suitable for dry skin”</i> recorded as suiting dry skin | all wording rules              |
| Universal claim expanded          | <i>“all skin types”</i> became Combination and Sensitive        | 751 products                   |
| Blank meant two things            | no evidence found, and question inapplicable, indistinguishable | lip and body products          |
| Claim stored without its sentence | tier recorded, evidence lost                                    | 963 quotes recovered           |
| One brand, five spellings         | per-brand grouping split; brand counts wrong                    | 104 brands                     |
| Placeholder strings as data       | coverage overstated                                             | all text columns               |
| Two shops disagreeing             | merge silently took the lower tier                              | 174 resolved by a third source |
| Resale listing treated as retail  | a private seller’s description carried a retailer’s tier        | 5 products demoted             |
| Markup swallowed by extraction    | CSS and usage text stored as ingredients                        | 1,550 formulas                 |
| Formula split on every comma      | <i>1,2-Hexanediol</i> became two ingredients                    | 2,775 cases                    |

Two classes recur and are worth separating.

*Rules that matched text without reading it.* Every wording rule searched for a positive phrase and none tested for negation, so *“not suitable for dry skin”* contains *“suitable for dry skin”*, the rule fired, and the product was recorded as suiting dry skin. The stored evidence quote contained the word *not*, which is how the fault was eventually found: the provenance column contradicted the value beside it. Separately, a source stating *“all skin types”* had been expanded into all five skin flags, and because dry and oily together imply Combination, 751 products carried a specific claim no source had made.

*Labels that meant more than one thing.* An empty skin-type field conflated *searched and found nothing* with *the question does not apply*, the second being true of lip balms. One manufacturer appeared under five spellings, affecting 104 brands and every per-brand figure.

### 4.3.1 Scope

Retailer categories are not scoped to skincare, and a listing is not always a product. 555 rows were removed in five groups, each written to a separate file so the exclusion is reproducible: colour cosmetics (196), offer listings such as buy-one-get-one (195), hair and hygiene products (80), accessories such as exfoliating gloves (59), and multi-item gift or travel sets (25).

Two exclusion rules were written and abandoned, and both are reported because both were plausible and neither produced an error. Treating *Duo* and *Trio* as set indicators would have deleted twelve La Roche-Posay Effaclar Duo products, which are a single formulation. Treating an ampersand as a sign of two products flagged 571 rows, among them *Vitamin C & Ferulic Acid Serum*.

## 4.4 Deduplication and cross-source matching

A product may be listed by several Lebanese shops and also appear in the international catalogue under a name written by its manufacturer. Establishing that two records describe one product is a record linkage problem rather than a lookup, and the procedure is reported here with the decision each stage required.

### 4.4.1 Blocking

Comparing every pair across three sources is quadratic and unnecessary. Candidate pairs were generated by blocking on the normalised brand, which reduces the comparison space by four orders of magnitude before any similarity is computed [81]. Normalisation strips case and punctuation, so variant spellings of one brand fall in the same block.

Duplicates within a single source are resolved first. A product stocked by four Lebanese shops becomes one row carrying four prices, rather than four rows competing to match the catalogue. The accepted risk is stated: a true match whose brand is spelled in a way normalisation does not repair is lost.

### 4.4.2 The two-metric acceptance rule

Matching uses token-based string similarity [30] under a rule requiring two metrics to agree: a pair is accepted only when `token_set_ratio` clears 90 *and* `token_sort_ratio` clears a lower floor.

**[TODO: State the numerical second-metric floor from the matching script, and confirm it against the accepted-pair distribution in the release. A rule reported without its threshold is not reproducible.]**

The two answer different questions, and neither is sufficient alone. `token_set_ratio` scores the shared subset of words and ignores words present in one name and absent from the other, which is what allows a retailer's decorated listing title to match a catalogue name. Because it ignores the extra words, it also scores a short string against a much longer one perfectly: *lip care* against *Beesline lip care shimmery strawberry* returns 100. `token_sort_ratio` normalises word order and penalises unmatched words, so it is sensitive to length and rejects that pair. Requiring both is what separates a decorated title for the same product from a different product sharing a few words.

Table 4.3 reports the seven measures considered on the same pair.

**Table 4.3.** Seven similarity measures scored on the same pair: *Foaming Cleanser* from the catalogue against *CeraVe Foaming Facial Cleanser Normal to Oily Skin* from a retailer. Only the token-set measure clears the acceptance threshold, and it is the reason the second metric is required.

| Measure                       | Score | Behaviour                                                                 |
|-------------------------------|-------|---------------------------------------------------------------------------|
| <code>token_set_ratio</code>  | 100   | Ignores words absent from the other string. Accepted, with a second gate  |
| <code>partial_ratio</code>    | 90    | Rejected: a substring scores 100 against a different product              |
| <code>WRatio</code>           | 86    | Weighted blend; less decisive                                             |
| <code>jaro_winkler</code>     | 65    | Built for short names, not product titles                                 |
| <code>token_sort_ratio</code> | 57    | Penalises unmatched words. Used as the second gate, and for deduplication |
| <code>ratio</code>            | 48    | Every extra word counts as many edits                                     |
| <code>QRatio</code>           | 48    | Inherits the same blind spot                                              |

#### 4.4.3 Products carried by two sources

519 products appear in both the international catalogue and Lebanese retail: 227 matched exactly and 292 by fuzzy matching at 90 or above, with the method and score retained for each. They are labelled *Both* rather than filed under either source.

These rows matter out of proportion to their number. They are the only records carrying an international listing and a Lebanese price for the same product, and therefore the only subset supporting any comparison between the two

markets. Field-level resolution is by source competence: the catalogue row supplies the formula and the aggregate rating, the retail row supplies the price and the stocking shop, and the merged row records that both contributed.

## 4.5 Ingredient recovery

Ingredient coverage began at 83.8% and reached its present level through seven routes. Their individual yields are more useful to anyone building a comparable resource than the final figure: open web search reached 23% of the products it was given, a third-party ingredient database 10%, and a crawl of manufacturer sitemaps 4%.

Seven page layouts had to be handled: a specification table row, a meta description, a block of capitalised names under a *Composition INCI* heading, running text after a heading, a line under a second heading, a bulleted list containing no commas, and a formula with no heading near it. The last three were found only after routes written for the first four reported those brands as publishing nothing.

Four defects in the acceptance rules were rejecting genuine formulas, and each produced output indistinguishable from a manufacturer that publishes nothing. The shape test required a formula to open with a water-phase ingredient, which discarded every anhydrous product: sticks, balms, oils and saponified soaps. Headings were matched in English only, so *Ingrédients* and *Composition* were invisible. The token *inci* matched inside words such as *principal*. And bullet separators were not normalised, so a formula written with bullets contained no commas and failed the first condition. All four were found by printing, for every page matched but not extracted, whether ingredient wording appeared and in what layout.

## 4.6 Regulatory linkage

The step that changes what the dataset can claim is the link from each ingredient to its entry in CosIng, the European Commission’s inventory maintained under Regulation (EC) No 1223/2009 [38, 40]. Before it, a screening rule flagging a product as possibly drying rested entirely on a list written for this project. After it, the registry contributes a citable fact alongside the rule: the product contains

ALCOHOL DENAT., which the Commission records under register entry 74174 with the functions astringent and solvent.

The distinction matters and is easy to overstate. A recognised function is registry data about a substance. It does not establish that a finished formulation, at its actual concentration and in its actual vehicle, worsens dry skin. The registry strengthens the provenance of the screening rule's input; the rule itself remains a project heuristic and carries its own evidence grade.

Matching is exact by design. Names are compared after lowercasing and removing brackets, because a shop writes *Aqua (Water)* where the register writes AQUA, and nothing fuzzy is permitted beyond that. A near miss on a chemical name is a different chemical, and a wrong ingredient would quietly produce a wrong safety claim. Reaching the present rate required repairing the stored strings as well as improving the lookup: splitting on every comma had broken every INCI name containing one, and the splitter now protects commas between digits and inside brackets.

The denominator must be stated with the rate, because three parsers give three different totals over the same file. The stored `ingredient_count` column sums to 295,991. The pipeline's own splitter, which protects commas inside numbers and brackets, yields 296,512. A naive split on every comma yields 299,351. The figure reported here uses the stored count, which is the smallest of the three and therefore the least favourable choice available: the two larger denominators would lower the rate to 96.4 % and 95.5 % respectively.

The stored count is chosen because it is the column the pipeline wrote beside `cosing_matched`, not because it flatters the result. That choice needs one qualification. In 80 records `cosing_matched` exceeds the stored `ingredient_count`, by 998 mentions in total and by as many as 88 in a single record, which is impossible if both columns count the same units over the same string. The two columns are therefore not yet demonstrated to share a counting basis.

**[TODO: Recompute `ingredient_count` and `cosing_matched` in one pass with one parser and report the rate from that single computation. Until then the 80 records above are an open question about the unit, not a rounding artefact, and the rate below should be read as provisional.]**

On that basis 285,917 of 295,991 mentions resolve against the register, a mention-level rate of 96.6%. At product level, 11,704 of the 11,802 products holding a formula carry at least one matched ingredient, the median share of a formula matched is 100 %, 10,288 formulas match at ninety per cent or better and

9,032 match completely. All figures in this chapter are computed from the release identified in Section 4.13; a statistic from any other release is not comparable.

These are coverage figures and not accuracy figures. What they establish is the proportion of ingredient strings that resolve to a register entry, not that each resolution names the substance the manufacturer intended. Exact matching after normalisation can still lose a chemically meaningful distinction, and the correctness of a sample of resolutions has not yet been measured. That measurement is specified in Chapter 8. Four fields record the outcome: the functions the Commission recognises for the ingredients present, the number matched, that number as a share, and any substance the register restricts.

9,613 products name at least one restricted substance, and that figure is not reported as a warning. Phenoxyethanol is restricted by concentration and is an ordinary preservative; the ultraviolet filters in a sunscreen are restricted under Annex VI and are the reason the product works. The field allows a user to ask the question rather than answering it for them.

*What is in the released file, and what is not.* The released columns carry the recognised functions present in a formula, the count matched, the coverage share and the names of restricted substances. They do not yet carry a per-ingredient mapping to a CosIng identifier or to an annex entry. Publishing that mapping is what would make every regulatory reference recoverable from the dataset alone, and it is listed in Chapter 9 as the next release step rather than claimed here. CosIng is the source from which CosIng-KG is built [23], so the alignment is available; it is not yet materialised in the file.

## 4.7 Field inventory

The dataset began with 42 fields describing identity, classification, text, market and reception. It now carries 111. The additional fields are not new observations: no site was scraped a second time and no external service was queried to produce them. They are derived from the ingredient list already present on each record, by the rules of Section 4.8.

**Table 4.4.** Field inventory and coverage over the 12,629 products. Groups one to five are observed; group six is the link to the European register; groups seven to ten are derived from the ingredient list by the rules of Section 4.8. Coverage is the share of rows carrying a non-empty value. Derived fields are bounded above by the 11,802 products that publish a formula.

| Field                              | Populated | Coverage |
|------------------------------------|-----------|----------|
| <i>Identity and classification</i> |           |          |
| product_id                         | 12,629    | 100.0 %  |
| source_category                    | 12,629    | 100.0 %  |
| brand                              | 12,629    | 100.0 %  |
| name                               | 12,629    | 100.0 %  |
| product_type                       | 12,629    | 100.0 %  |
| country                            | 12,629    | 100.0 %  |
| product_url                        | 12,610    | 99.8 %   |
| <i>Skin suitability</i>            |           |          |
| skin_type                          | 12,629    | 100.0 %  |
| sensitivity                        | 12,629    | 100.0 %  |
| skin_type_source                   | 12,629    | 100.0 %  |
| <i>Formulation, as published</i>   |           |          |
| ingredients                        | 11,802    | 93.5 %   |
| ingredient_count                   | 11,802    | 93.5 %   |
| key_ingredients                    | 11,129    | 88.1 %   |
| free_from                          | 10,963    | 86.8 %   |
| spf                                | 1,378     | 10.9 %   |
| benefits                           | 12,220    | 96.8 %   |
| concerns                           | 11,802    | 93.5 %   |
| <i>Market</i>                      |           |          |
| price_usd                          | 11,700    | 92.6 %   |
| price_lbp                          | 11,700    | 92.6 %   |
| price_seen_date                    | 5,469     | 43.3 %   |
| shops_in_lebanon                   | 11,937    | 94.5 %   |
| sold_by_shops                      | 5,643     | 44.7 %   |
| price_per_ml                       | 3,387     | 26.8 %   |
| size_ml                            | 3,413     | 27.0 %   |
| <i>Reception</i>                   |           |          |
| rating                             | 6,487     | 51.4 %   |
| rating_count                       | 6,487     | 51.4 %   |
| rating_source                      | 6,963     | 55.1 %   |
| review_texts_json                  | 4,764     | 37.7 %   |
| <i>Regulatory link (CosIng)</i>    |           |          |
| ingredient_functions               | 11,703    | 92.7 %   |
| cosing_matched                     | 11,704    | 92.7 %   |
| cosing_coverage                    | 11,704    | 92.7 %   |
| restricted_ingredients             | 9,613     | 76.1 %   |
| <i>Derived: regulatory watch</i>   |           |          |
| contains_nanomaterial              | 11,802    | 93.5 %   |
| contains_solid_microplastic        | 11,802    | 93.5 %   |
| contains_synthetic_polymer         | 11,802    | 93.5 %   |
| contains_endocrine_concern         | 11,802    | 93.5 %   |
| contains_pfas                      | 11,802    | 93.5 %   |
| <i>Derived: actives and risk</i>   |           |          |

## 4.8 Derived attributes and evidence

### 4.8.1 Three kinds of derivation

Derived fields do not all come from the same input, and treating them as though they did would misstate what bounds their coverage.

*Ingredient-based* fields are read from the INCI list and are bounded above by the 11,802 products that publish one. They are the majority and the subject of the rest of this section. *Text-based* fields are read from the product description rather than the formula: `texture` is assigned from descriptive wording, which is why Section 4.8.3 separates its vocabulary from its measurement. *Metadata-based* fields are computed from columns that are neither formula nor description: `routine_step` and `body_site` from product type, `regional_style` from country of origin, `price_band_lebanon` from the observed price, and `brand_parent_company` from brand. These last are present on records with no formula at all, which is why their coverage exceeds 11,802.

The ingredient-based rules rest on two properties of the INCI list, and both are legal requirements.

Article 19(1)(g) of Regulation (EC) No 1223/2009 requires ingredients to be listed in descending order of weight at the time of addition, and permits ingredients below one per cent to appear in any order [40]. Order above the one per cent mark is therefore information rather than presentation. What it does not carry is a concentration. The position of the one per cent boundary in a given list is unknown, the minimum effective concentration of a substance is not recoverable from a label, and an active at position four is ranked above one at position thirty-four without either being quantified. `lead_active_position` is accordingly an *ordinal* feature, recorded for the 4,376 products carrying an identifiable active, and it is used for ranking rather than read as a dose. The same article requires a nanomaterial to carry the word (*nano*), so detecting that bracket detects a declaration made under law.

Each rule is a list of INCI marker strings matched against the split ingredient list, with an exclusion list where a marker would otherwise produce a known false positive. A match writes the flag and, for most rules, the *first* matching ingredient string into a companion evidence field. Two limitations follow and are carried into the release plan of Section 4.13: only the first match is retained rather than all of them, and several flags have no companion evidence column

at all. A reproducible explanation needs every match, the rule identifier and its version, and the source of the formula, which the present release does not supply.

### 4.8.2 The evidence grade

Derived fields do not rest on evidence of equal quality. Each is graded, and the grade is carried into the ontology as a property of the claim.

*The four grades.* **A, law.** The rule follows from a regulation with an article number. **B, published science.** The rule follows from a peer-reviewed result, open to disagreement about whether the finding transfers to the concentrations used in a marketed product. **C, industry practice.** The rule is what cosmetic analysis services apply, with no single publication behind it. **D, presentation label.** The rule was defined for this project and is not defended as a finding.

### 4.8.3 The rules, in full

Table 4.5 states, for every derived field, the markers that trigger it, the authority for the rule, its grade and the count. Six rules need more than a table cell, either because the marker list embeds a judgement or because the underlying evidence is contested.

*Comedogenicity.* Markers: isopropyl myristate, isopropyl palmitate, cocos nucifera oil, lauric acid, myristyl myristate, octyl palmitate, ethylhexyl palmitate, linseed oil, wheat germ, laureth-4, oleyl alcohol, sodium lauryl sulfate, butyl stearate, isocetyl stearate, myristyl lactate, cocoa butter, theobroma cacao. Seventeen markers, flagging 1,991 products.

The list is the one cosmetic analysis services apply, and it descends from the rabbit ear assay: a candidate substance is applied inside the ear of an albino rabbit for two to four weeks and the resulting comedones are counted. The published ingredient tables still in circulation come from that work [44]. The same investigator later tested the same substances on human subjects and found that strong comedogens in the rabbit produced little or no response in people, and the concept was re-evaluated in the clinical literature on that basis [35]; the regulatory gap was restated recently [6]. The field is retained because consumers look for it and is reported as a screening heuristic of contested provenance, never as a prediction.

**Table 4.5.** Derived rule families, what licenses each, and its grade. A a regulation with an article number; B a peer-reviewed result; C industry practice, or a screening flag under a proposal or a review, named as such; D a label defined for this project. B/D marks a rule whose general principle is published but whose operational form — the marker list, the number of levels, the keyword mapping — was defined here and is not itself validated. The *Primary fields* column counts the fields each family produces and reconciles to the column total below it; companion evidence fields and observed fields are listed separately rather than folded in. The complete registry is Appendix C.

| Rule family                       | What licenses the rule                                                                                                                                                    | Grade | Primary fields |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------|
| Label declarations                | Art. 19(1)(g): the ( <i>nano</i> ) marker and descending-weight order [40]                                                                                                | A     | 2              |
| Enacted restrictions              | Annexes to Reg. (EC) 1223/2009 via CosIng; Reg. (EU) 2023/2055 solid particles [3, 38]                                                                                    | A     | 3              |
| Screening under review            | Substances under SCCS review; proposed PFAS restriction. Not enacted law                                                                                                  | C     | 2              |
| Barrier and lipid                 | Lipid-mixture experiments [70, 108]; niacinamide separately [95]. The twelve-marker list is unvalidated                                                                   | B/D   | 2              |
| Photostability and use            | Retinoid degradation under UV [2]                                                                                                                                         | B     | 1              |
| Tolerance and sequence            | Retinoid irritancy ordering and escalation [13, 43]; sequential-application effects [27]. The eleven steps, three levels and warning list are this project's              | B/D   | 4              |
| Microbial lipid feeding           | <i>Malassezia</i> lipid dependence [7]. Which cosmetic esters matter in use is unsettled                                                                                  | B/D   | 1              |
| Sensory vocabulary                | Instrumental-sensory correlation [10, 48]. Assigned here from product text, not measured                                                                                  | B/D   | 1              |
| Comedogenicity                    | Rabbit-ear assay [44], contested in human subjects [6, 35]                                                                                                                | C     | 1              |
| Analysis-service vocabulary       | What cosmetic analysis services flag: acids, alcohols, fragrance, filters, animal origin                                                                                  | C     | 9              |
| Formulation convention            | Water phase, preservative family, humectant against occlusive                                                                                                             | C     | 4              |
| Presentation labels               | Defined for this project; not defended as findings                                                                                                                        | D     | 3              |
| Presence filters                  | The <i>free-from</i> vocabulary consumers expect: PEG, talc, silicone, sulfate, propylene glycol, tree nut, gluten, soy, coconut and the remaining single-substance flags | C     | 22             |
| <b>Primary derived fields</b>     |                                                                                                                                                                           |       | <b>55</b>      |
| Companion <i>_evidence</i> fields |                                                                                                                                                                           |       | 18             |
| Observed fields, not derived      |                                                                                                                                                                           |       | 38             |
| <b>Columns in the release</b>     |                                                                                                                                                                           |       | <b>111</b>     |

*Malassezia-feeding ingredients.* Markers: polysorbates, sorbitan esters, glyceryl stearate, the free fatty acids from lauric through stearic and linoleic, caprylic/capric esters, the PEG stearates, laurates and oleates, and the ferment-derived ingredients galactomyces, saccharomyces and lactobacillus. Flags 7,093 products.

*Malassezia* species lack the genes for fatty acid synthesis and must take fatty acids from their environment, with growth documented on chains of roughly eleven to twenty-four carbons [7]. The mechanism is published; which cosmetic ingredients matter at the concentrations found in a marketed formula is not settled. At 56.2% of the catalogue the field is too broad to function as a filter, and it is surfaced as an ingredient some users avoid.

*Barrier support.* Markers: ceramide, cholesterol, phytosphingosine, sphingolipid, niacinamide, panthenol, centella, madecassoside, squalane, linoleic acid, beta-glucan, allantoin. Flags 5,690 products.

The stratum corneum barrier is held together by ceramides, cholesterol and free fatty acids. Man et al. [70] showed that applying an incomplete subset of the three delays recovery and that recovery accelerates when one is raised up to threefold; the result was extended to chronologically aged skin by Zettersten et al. [108]. The marker list is wider than those three lipids because it also counts soothing agents, so the field is more useful as a filter and less defensible as a claim than a strict variant would be. Both are reported in Chapter 8.

*Retinoids, night use and layering.* Markers for the retinoid family: retinol, retinyl, retinal, retinoic, tretinoin, adapalene, retinaldehyde. 681 products.

`use_time` is set to night for products carrying a retinoid or benzoyl peroxide, because retinoid compounds isomerise and oxidise under ultraviolet exposure, a photostability property of the molecule [2]. `do_not_layer_with` records the conflict between a retinoid and an acid or ascorbic acid in the same application, which rests on the same sequential-application evidence as routine position [14, 27]. `may_cause_purging` covers retinoids together with salicylic, glycolic, lactic, mandelic and azelaic acids and benzoyl peroxide; the temporary worsening these produce is described in the clinical literature as an expected and self-limiting reaction rather than an adverse event [13].

*Exfoliating acids, and the exclusion that matters.* Markers: glycolic, lactic, mandelic, malic and tartaric acid for alpha hydroxy acids; salicylic acid, betaine salicylate and capryloyl salicylic for beta hydroxy acids;

gluconolactone and lactobionic acid for poly hydroxy acids. 1,254, 1,026 and 409 products.

Citric acid is deliberately excluded. It is an alpha hydroxy acid by definition and is present in most formulas that contain it as a pH adjuster, at a level that does not exfoliate. Including it would flag 2,842 additional products as exfoliants.

*Regulatory watch, and the solid-particle distinction.* `contains_solid_microplastic` matches `nylon-n`, `polyethylene`, `polypropylene`, `polymethyl methacrylate`, `polyurethane`, the styrene/acrylates copolymers and PTFE: 589 products. `contains_synthetic_polymer` adds `carbomer`, `acrylates copolymer`, `polyquaternium` and the crosspolymers, and is reported separately and informationally: 3,166 products. The separation follows the instrument. Regulation (EU) 2023/2055 restricts *solid* synthetic polymer microparticles [3], and carbomer and the polyquaterniums are dissolved or swollen polymers rather than solid beads.

`contains_endocrine_concern` matches substances under active review by the Scientific Committee on Consumer Safety, including butyl- and propylparaben, benzophenone-3, triclosan, butylated hydroxyanisole, the cyclosiloxanes and resorcinol: 1,582 products.

#### 4.8.4 Two fields raised from convention to evidence

*Routine position.* `routine_step` maps product type to a position in an eleven-step sequence. Sequential application alters percutaneous absorption: Burstein et al. [27] review twelve studies and attribute the effect to barrier modification and vehicle interaction rather than to formulation class, and Beebeejaun et al. [21] measure roughly a fivefold difference in the absorption of one corticosteroid according to which product was applied first. Ahuja et al. [9] report that no formal guideline fixes the order and Arce et al. [14] find the same effect for a cosmetic active applied in layers. The step numbers remain a convention of this project; the claim that order matters does not.

*Strength level.* `strength_level` assigns beginner, intermediate or advanced from which actives are present and at what position. Fluhr et al. [43] order the retinoid family by irritancy across forty-four weeks in 355 participants, Mellody et al. [71] report a six-week escalation regimen in which 0.3 % retinol was better tolerated than 1 % at comparable effect, Aneri et al. [13] record Delphi agreement among 133 dermatologists that gradual and spaced introduction at night is appro-

prate, and Vashkevich et al. [100] describe the adaptation phase. The three-level granularity is this project's.

### 4.8.5 Fields that remain presentation labels

`regional_style` maps country of origin to a market tradition and describes the market usefully without any published classification behind it. `price_band_lebanon` divides the catalogue into bands whose boundaries are chosen here, although the prices themselves are original observations carrying a currency, and a date for the catalogue segment only.

`body_site` is treated differently. Its values are a short anatomical list present on every record: face for 10,704 products, body for 676, eye area for 637, lips for 550 and hands for 62. Each value is aligned to a term in UBERON [75], so a query for products applied to a facial site can be expressed without this project's vocabulary being the authority. The alignment is listed in Appendix B.

*What a negative value means.* Each flag takes three states and they must not be collapsed. *Yes* means a marker from the rule's list was found, and the evidence field names the ingredient that matched. *No* means no marker from that list was detected in the formula as held, which is weaker than confirmed absence: the list is finite, the stored formula may be incomplete, and a substance under an unlisted synonym would not match. *Blank* means no formula is held for that product at all, and covers the 827 records without one.

The consequence is asymmetric and is stated rather than smoothed over. This resource can establish that a product contains something. It cannot establish that a product is free of it, which is why no certification-style claim such as vegan status is derived.

### 4.8.6 What could not be derived

Cruelty-free status is a fact about a company's testing process, organic status a fact about how a plant was farmed, and vegan status requires proving an absence. None is a property of a molecule and no ingredient list carries any of them. Text mining was attempted first and failed separately: the product summary field proved to be generated boilerplate, and searching it for a certification phrase returned sixteen matches across the whole catalogue. All three are properly modelled as a claim made by a named certifier on a date, a shape Chapter 5 can express and this dataset does not populate.

## 4.9 Coverage and provenance

### 4.9.1 Provenance, and the three counting rules

Every suitability claim records the source that asserted it, and the counting rule has to be named because two measurements of the same thing disagree by a known amount.

Counting by source string, 1,624 product records carry a `skin_type_source` value containing the word *inferred*. Counting by evidence tier, 1,676 records sit in tier four, the tier reserved for a value inferred from the formula. The difference of 52 records is not a discrepancy in the data but a labelling gap: those 52 are assigned tier four while their source string does not use the word, and they are listed in the release anomaly report.

The tier-based count is the one used in this thesis. Tier one, the manufacturer, holds 4,460 records; tier two, a retailer, 4,539; tier three, a weaker published source, 1,954. Tiers one to three therefore hold 10,953 records, or 86.7% of 12,629. A third, status-based count separates 215 *not applicable* records, which are neither stated nor inferred, so the complement of the inferred count should not be read as proof that every remaining value was stated by a source.

One further qualification applies throughout. These are *product records carrying an evidence tier for the skin-type field*, not individual claims. A product may carry several suitability statements and the tier attaches to the record, so a claim-level provenance count would be a different measurement and has not been made.

Supporting text is held for a substantial but not complete share of claims: 10,351 rows retain the sentence the claim was read from and 11,404 the page it appeared on. The tier is what makes the merge rule of Section 4.3 possible, and it becomes the evidence level in the ontology.

### 4.9.2 The market layer, and what it does not establish

11,700 products carry a price in United States dollars. The Lebanese pound figure beside it is a conversion of that price at a single stored rate of 89,500 to the dollar, not a second independent observation, and it is described as a conversion rather than as a price in a second currency.

Availability is recorded as listing rather than as stock. `shops_in_lebanon` is populated for 11,937 products, but the value is zero for the 6,294 catalogue products that no Lebanese shop lists; 5,643 products have a positive count and

356 are listed by more than one shop. A populated field and a positive count are different statistics, and only the second is evidence of local availability.

*The limitation this release carries, stated plainly.* 5,469 products carry a price observation date. Of those, 5,453 are catalogue products whose price is international; only 10 Lebanese retail products and 6 Lebanese origin products carry a date, and none of the 519 products does. Dated *local* price coverage is therefore close to absent.

This thesis argues that a price is meaningful only with a currency and a date, and the Lebanese layer of the present release does not meet that standard. It records that a seller listed a product at a price, without recording when. The listing is evidence of availability at collection time and the collection window is documented, but a per-product observation date for local prices is future work rather than a delivered property, and the claims of Chapter 7 about price freshness are bounded accordingly.

Reception data is thinner still and is reported as such: 6,487 products carry a rating and 4,764 carry review text. No user identity and no interaction history exists anywhere in the resource, which is the scope condition of Section 1.5 and the reason the evaluation of Chapter 8 takes the form it does.

## 4.10 Technical validation

### 4.10.1 Automated checks

A validation suite verifies that the dataset and its evidence file describe the same products with one row each and no duplicate identifier; that the stored ingredient count agrees with the formula under the same splitting rule; that no markup survived extraction; that no field derived from a formula is set on a row without one; that prices, ratings and coverages lie inside their possible ranges; and that excluded categories did not reappear.

Three checks fail on the current release and are reported rather than suppressed. The controlled-value check on `source_category` rejects *Global* and *Both* because the allowed-value list predates the introduction of the overlap label, so the check is out of date rather than the data. The ingredient-count check disagrees with the stored count on 2,982 records under the validator’s own splitting rule; under the pipeline’s protected-delimiter parser the disagreement falls to 252 records. The two figures are results of different parsing procedures and are not

interchangeable, which is why the denominator in Section 4.6 is named explicitly. The `free_from` check flags 516 records whose stored ingredient count falls below eight. That threshold is the script’s heuristic for a formula that may be truncated; it does not establish that any of the 516 is incomplete. A two-ingredient olive oil soap and a truncated extraction both fall below it. These are records flagged for inspection, and separating genuinely short formulas from extraction failures and from unsupported free-from claims is work still to be done.

Separately, six anomaly classes are counted and published with the release: 52 tier-four rows whose source string does not carry the word *inferred*, which accounts exactly for the discrepancy reconciled in Section 4.9.1; 155 products classed anhydrous whose formula names water at a later position; 67 products where the matched-ingredient count exceeds the token count under the protected splitter; and the undated prices already reported. Naming the failures and the anomaly counts is what allows a reader to judge the release; a suite reported as wholly passing would not.

Three of the checks were themselves wrong when first written, and the distinction between a failing check and a failing dataset had to be made by inspection. The count check split *1,2-Hexanediol* into two ingredients. The short-formula check flagged 261 legitimate single-ingredient products, among them 29 sugar scrubs whose entire declared composition is *Sucrose*. The scope check flagged a mascara *remover* as colour cosmetics. Had those failures been trusted, correct data would have been deleted to satisfy them.

### 4.10.2 Measured error rate on third-party matching

Formulas obtained by matching a product name against a third-party catalogue carry the weakest evidence tier, and their error rate was measured rather than assumed. All 27 recoveries from the ingredient-database route were read by hand against the matched record: five were a different product and four more were doubtful, giving an error rate between 19 and 33 per cent. The five confirmed errors were removed and the four doubtful cases retained and named in the repository.

That measurement is the justification for the tier system rather than a footnote to it. During development a general web retailer matched an anti-acne serum to a men’s fragrance and a sunscreen to a body lotion, extracting a clean, plausible and entirely wrong formula in both cases. That source was excluded on

accuracy grounds, and a match now additionally requires the brand to appear in the matched URL.

### 4.10.3 A cross-check between two independent layers

Because suitability claims carry provenance and ingredients carry regulatory status, the two can be checked against each other. 2,223 products declare at least one of the twenty-six fragrance allergens the European Union then required to be labelled by name, and 587 of those are marked as suitable for sensitive skin.

These 587 are cases requiring further assessment, not demonstrated errors. A declarable allergen is declared above a concentration threshold, and a product containing one may still be suitable for a user whose sensitivity is unrelated to that substance. What the two recorded statements establish is that a suitability claim and a regulated ingredient coexist on the same product, which is a case a reviewer should look at. The dataset records both rather than overruling either, and that is only possible because the claim retains its source.

### 4.10.4 Rule validation

Each derived rule was validated by drawing random matching products and reading the matched ingredient against the rule. Four rules were wrong. Citric acid was counted as an exfoliating acid, which would have flagged 2,842 products incorrectly. Carbomer was counted as a solid microplastic, accounting for 1,930 of 3,125 initial hits and producing the split reported in Section 4.8.3. Cetearyl alcohol was counted as a drying alcohol when fatty alcohols are conditioning, reducing that count from 1,503 to 826. Synthetic beeswax was counted as animal-derived, affecting 43 products.

## 4.11 Limitations

**The missing formulas are not random.** They concentrate in brands whose sites refuse automated access, brands with no reachable website, and simple artisanal products whose entire composition is two or three ingredients written as prose. The shape test does not accept formulas of fewer than five ingredients, which excludes a small number of genuinely simple products such as a two-ingredient olive oil soap.

**Two sources were deliberately not used.** One ingredient database and two large brand sites return HTTP 403 to automated requests. Circumventing an

explicit refusal was rejected as a method; thirty-one formulas from one of those brands were instead read by hand from pages open to any reader.

**Size is present for 27.4 % of products** and cannot be improved from these sources: of the products without a size, only 256 state one anywhere in any field. **Ratings are present for roughly half.** Many Lebanese shops run storefronts with no review application installed, so their pages carry no review data at any depth of reading.

**Prices are a snapshot, and most local prices are undated.** Lebanese prices move quickly, which is exactly why the missing observation dates matter: 5,453 catalogue prices carry a date and only 16 local ones do (Section 4.9.2). Prices for catalogue-only products are international rather than Lebanese, and a separate field distinguishes them.

## 4.12 Usage conditions

*What a consumer of this dataset must do.* Filter on the evidence tier to control claim strength: tiers one to three are claims made by an identifiable source, tier four is inferred from the formula, and a recommender treating the two identically will make confident statements it cannot support. Filter on source category for availability: *Global* products are not known to be sold in Lebanon, and *Both* products carry an international listing and a local listing with a price, though not an observation date. Treat a third-party ingredient list as weaker than a manufacturer's, at the measured error rate of Section 4.10.2. Do not read an empty restricted-ingredient field as missing data: it means no restricted substance was found in the formula as held. And read a negative derived flag as *no listed marker detected*, never as confirmed absence.

## 4.13 Publication

The dataset, the exclusion files, the validation suite and the collection scripts are published in the project repository. Data were collected from publicly published product pages for academic research.

**[TODO: State the repository URL, the exact release tag and the commit hash from which every figure in this chapter was computed, and confirm the repository is publicly reachable. The URL previously given resolved to a 404 at review, so the identifier must be verified**

**before submission. Regenerate the statistics and the validation report from that one release so the thesis, the portal and the data descriptor cannot disagree.]** Each script documents in its opening comment why it exists and, where applicable, the defect in an earlier version that motivated it, so a field declared a presentation label in Section 4.8.3 can be read back to its definition.

Preparing the repository required removing credentials from the collection scripts, excluding intermediate files and cached responses, and recomputing every figure in the accompanying documentation from the published file rather than copying it forward. This chapter follows the same discipline: every number in it is held in a macro recomputed from the dataset, so the prose cannot drift from the data. Prices are released in dollars with a stored conversion rate; an observation date accompanies the catalogue segment and, as Section 4.9.2 reports, almost none of the local records.

**[TODO: Add the archival DOI once the deposit is minted, and the licence under which the dataset is released.]**

## Chapter 5

# Ontology Design and Construction

### 5.1 Methodology

[TODO:MOMo for design, LOT for requirements, publication and maintenance. One paragraph on why each, with the four reuse-failure modes you actually hit as the justification. Do not restate Chapter 3.]

### 5.2 Requirements specification

#### 5.2.1 Purpose, scope and intended users

#### 5.2.2 Competency questions

[TODO:The full list goes in Appendix A; give the grouped summary here and the rationale that every question is answerable from a field the dataset already holds.]

### 5.3 Vocabulary reuse

[TODO:The decision table: what was reused, what was referenced by MIREOT, what was rejected and why. The full audit goes in Appendix B.] [note:Keep the *decisions* here and the *descriptions* in Chapter 3. A reader should never have to ask why your choice appears in your literature review.]

### 5.4 Conceptualisation

#### 5.4.1 Module structure

[TODO:The modules and why they are separated: they change at different rates. Include the schema diagram for each module.]

### 5.4.2 Core classes and properties

### 5.4.3 Defined classes

[TODO: Each equivalence axiom, in Turtle, with the count of products the reasoner places in it and whether that count matched the prediction.]

## 5.5 Formalisation

[TODO: Turtle listings for the axioms that matter. Do not paste the whole file; it goes in the repository.]

## 5.6 Publication

[TODO: Permanent identifier, generated documentation, archived release with a DOI, versioning and maintenance policy. This is a contribution in a field where one artefact in four is resolvable, so report it as one.]

## Chapter 6

# Knowledge Graph Construction and Enrichment

### 6.1 Declarative population

[TODO:The mapping language, the mapping file, and why a declarative mapping rather than a script: it is publishable, reviewable and re-executable.]

#### 6.1.1 Named graphs by source

[TODO:One named graph per source, so a source can be reloaded alone and a query can be restricted to sources the user trusts.]

#### 6.1.2 Resulting graph

[TODO:Triple counts by class and by named graph.]

### 6.2 Validation

[TODO:SHACL shapes mirroring the tabular validation suite, and what the reasoner found that the tabular checks could not express.]

### 6.3 Enrichment from text

#### 6.3.1 What is extracted, and from what

[TODO:Product descriptions and review texts. State coverage honestly: review texts are present for a minority of products.]

### 6.3.2 Method

[TODO: The NLP method, with the validation rule: every proposed term is checked against an authoritative source before it enters the graph.]

### 6.3.3 What survived validation

[TODO: The survival rate is the result. Report it even if it is low.]

## 6.4 Discussion

[TODO: What enrichment added that construction could not.]

## Chapter 7

# Reasoning-Based Recommendation and Ingredient Similarity

## 7.1 Recommendation as constrained selection

[TODO:Frame the task as constraint satisfaction rather than ranking, and say why: an allergen exclusion is a filter, not a preference to weight.]

## 7.2 Semantic reasoning

### 7.2.1 Defined classes and inferred membership

### 7.2.2 Rules that OWL cannot express

[TODO:Where rules were needed, with their clinical source cited.]

### 7.2.3 Explanation

[TODO:The path from user need to recommended product, and how it is rendered as a reason the user can read.]

## 7.3 Ingredient-based similarity

### 7.3.1 Representing a formula

[TODO:INCI lists are ordered by descending concentration above one per cent, so position carries information. Say how you use it.]

### 7.3.2 Choice of similarity measure

[TODO: The comparison, with the worked example and the measured numbers.]

### 7.3.3 Substitution

[TODO: The step nobody else can take: when the suitable product is not available locally, an ingredient-similar product that is available becomes the recommendation, and the shared composition is the explanation.]

## 7.4 Worked examples

[TODO: Two or three real users, end to end, with real products and real Beirut prices. This is the section a supervisor will read first.]

## Part III

# Evaluation and Discussion



## Chapter 8

# Evaluation

### 8.1 Evaluation design

[TODO: State the objective, then the method, then the measures — in that order, following an established evaluation framework. Say plainly that accuracy-based recommender metrics do not apply in the absence of interaction data, and why that is a scope condition rather than a gap.]

### 8.2 Structural evaluation

[TODO: Pitfall scanning and FAIR scoring. Report the scores, fix what is cheap to fix, and report the rest.]

### 8.3 Functional evaluation

[TODO: Every competency question answered by a query. Table of question, query, result count. This is the core of the chapter.]

### 8.4 Logical evaluation

[TODO: Predict how many products should fall into each defined class, then run the reasoner and compare. Treat every inconsistency as a finding.]

## 8.5 Comparative evaluation

[TODO: The capability table from Chapter 3, now filled in from the delivered system rather than from intent.]

## 8.6 Expert review

[note: Optional and cheap: three or four pharmacists or dermatologists shown twenty recommendations and asked whether the stated reason is sound. That evaluates the reasoning rather than satisfaction, which is more than the reviewed systems achieved. Include only if you actually run it.]

## 8.7 Threats to validity

[TODO: Be specific and unprompted. An examiner who finds a threat you did not name will assume there are others.]

## Chapter 9

# Discussion and Conclusion

### 9.1 Answers to the research questions

[TODO: One subsection per research question, each answered in a sentence and then supported. If a question was not fully answered, say so here rather than leaving the reader to notice.]

### 9.2 Contributions revisited

[TODO: What was claimed in Chapter 1 and what was delivered, with the evidence.]

### 9.3 Limitations

[TODO: The honest list. Clinical review of the concern vocabulary, review coverage, and the absence of user interaction data.]

### 9.4 Future work

[TODO: The neuro-symbolic methods that this resource enables once a deployment produces interaction data, and the enrichment directions.]

### 9.5 Availability of artefacts

[TODO: Dataset, ontology, code, and the identifiers under which each is published. Two of eight reviewed artefacts are resolvable; make yours one of them and say so.]



Part IV

# Appendices



## Chapter A

# Competency Questions

**[TODO: The full list, grouped, each with the fields it draws on and the query that answers it in Chapter 8.]**

[note: Every question must be answerable from a field the dataset already holds. That is the design constraint, not a coincidence, and saying so is worth a sentence.]



## Chapter B

# Evidence Ledgers

This appendix holds two ledgers. The first backs every coded cell in the related-work comparison; the second records the vocabulary reuse decisions. They are separate because they answer different questions and because Table 3.3 depends on the first.

## B.1 Literature evidence ledger

Every cell in Table 3.3 is coded from a passage in the source publication. This ledger records, for each coded cell, the study, the capability as defined in the table, the code assigned, the page or section the code was read from, the supporting passage, and the interpretation applied where the passage does not use the table’s vocabulary.

**Table B.1.** The columns each ledger entry must carry. A code without the last three columns is an assertion rather than evidence, which is why the ledger and not the table is the primary record.

| Column         | What it records                                                                       |
|----------------|---------------------------------------------------------------------------------------|
| Study          | Citation key, resolving to the bibliography                                           |
| Capability     | Exactly as defined in Table 3.3                                                       |
| Code           | <b>P</b> , <b>o</b> , <b>X</b> , <b>–</b> or <b>U</b>                                 |
| Location       | Page, section or figure the code was read from                                        |
| Passage        | The supporting sentence, quoted                                                       |
| Interpretation | How the passage maps to the table’s vocabulary, where the source uses different terms |
| Audit status   | Whether the full text was read, or only the abstract and bibliographic record         |

## Ledger: Jannatin and Ratnasari [59]

Full text read. Fifteen cells, one per capability in Table 3.3, coded from the published article.

**Table B.2.** Evidence ledger for Jannatin and Ratnasari [59]. Audit status for every row: full text read. Locations are sections of the published article.

| Capability                      | Code | Location         | Passage and interpretation                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------|------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ingredients represented         | P    | §IV.A, Fig. 4(c) | <code>Ingredient</code> is one of the twelve top-level classes; the data properties <code>IngredientName</code> , <code>IngredientList</code> and <code>IngredientFunction</code> are declared. Direct representation, no interpretation required.                                                                                                                                 |
| Ingredient <i>order</i> used    | —    | —                | Position within the list is nowhere mentioned. <code>IngredientList</code> is declared as a data property with no statement that its order carries meaning. Coded as not documented rather than absent, since the publication does not establish absence.                                                                                                                          |
| Linked to a regulatory register | ○    | §IV.A, Fig. 4(c) | The data property <code>BPOMNumber</code> records registration with the Indonesian regulatory body, described as indicating “product registration”. This is a product-level attestation. No ingredient is linked to a register entry, and no annex, function or concentration limit is carried. Partial under this table’s definition, which requires linkage at ingredient level. |

*continued overleaf*

Table B.2 continued

| Capability                     | Code | Location | Passage and interpretation                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Restriction status represented | ○    | §IV.A    | <code>Allergen Type</code> is a top-level class; <code>excludeAllergenType</code> , <code>incompatibleWith</code> and <code>mayAggravateConcern</code> are declared, with an <code>InteractionWarnings</code> data property. Restriction is therefore represented, but the categories are the authors' own and no statutory instrument is cited behind any of them. |
| Skin type modelled             | P    | §III.B   | <code>Skin Type</code> is a top-level class, and the publication states that it was designed by “mapping ‘Skin Type’ to Fitzpatrick categories”.                                                                                                                                                                                                                    |
| Sensitivity as a separate axis | P    | §IV.B    | The worked example takes “Skin Type: Dry”, “Skin Concern: Acne” and “Sensitivity: Alcohol” as three separate inputs, with <code>userAvoids</code> and <code>excludeAllergenType</code> operating on the third. Sensitivity is therefore modelled independently of the dry–oily dimension, which is the criterion applied in note (a) of Table 3.3.                  |
| Source recorded per claim      | –    | —        | No property records which agent asserted a suitability, benefit or interaction. There is no analogue of <code>hasHeuristicSource</code> in Serna et al. [90]. Not documented.                                                                                                                                                                                       |
| Evidence strength graded       | –    | —        | No grading of any kind is declared. Not documented.                                                                                                                                                                                                                                                                                                                 |

*continued overleaf*

Table B.2 continued

| Capability                   | Code | Location | Passage and interpretation                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Country-level availability   | —    | —        | The catalogue is drawn from Indonesian platforms, but no property records the territory in which a product is available. Not documented.                                                                                                                                                                                                                                                                              |
| Seller-level product listing | —    | —        | <b>Brand</b> and <b>Product Category</b> are represented; no retailer, shop or listing entity appears in the class hierarchy. Not documented.                                                                                                                                                                                                                                                                         |
| Price recorded               | —    | —        | No price property appears among the eighteen data properties in Fig. 4(c). Not documented.                                                                                                                                                                                                                                                                                                                            |
| Price observation date       | —    | —        | No temporal property of any kind is declared. Not documented.                                                                                                                                                                                                                                                                                                                                                         |
| Reasoner applied             | P    | §IV.B    | “The recommendation engine was implemented using Apache Jena Fuseki as the primary reasoning engine”, with inference over <code>suitableForSkinType</code> , <code>treatsConcern</code> , <code>incompatibleWith</code> and <code>avoidFor</code> . Coded <b>P</b> ; note that no OWL profile and no description-logic construct is named, so what is demonstrated is multi-hop traversal rather than classification. |

*continued overleaf*

Table B.2 continued

| Capability             | Code | Location | Passage and interpretation                                                                                                                                                                                                                                        |
|------------------------|------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Explanation produced   | –    | §V       | Explainability is claimed in the discussion, but no explanation artefact is shown, described or evaluated, and the interface figure shows a product listing only. A claim in the discussion is not a demonstrated capability, so this is coded as not documented. |
| Alternatives retrieved | –    | §IV.A    | A <code>similarTo</code> object property is declared, but no substitution or alternative-retrieval function is described or demonstrated in the system. Declared vocabulary without a demonstrated function is coded as not documented.                           |

[TODO:Populate this ledger for all eleven capabilities across the eight studies. Priority order, because these carry the novelty argument: the regulatory-register link, seller-level availability, the price observation date, and per-claim provenance. Jannatin and Ratnasari [59] is complete above. Utari et al. [99] is coded U throughout and its rows cannot be completed until the full text is obtained; record that as the audit status rather than leaving the cells blank. Until this ledger exists, Table 3.3 rests on codings a reader cannot check, and the gap statement in Section 3.10 inherits that weakness.]

## B.2 Vocabulary reuse audit

[TODO:The full audit: every candidate vocabulary, what it would serve, the decision taken, and the basis for it. The summary goes in Section 5.3; the reasoning goes here. Record rejections with their reasons: a vocabulary left out for a stated reason is a stronger position than one imported to look thorough.]



## Chapter C

# Data Dictionary

[TODO: All 111 columns: name, type, coverage, source, derivation rule. Generate this from the CSV with a script rather than typing it, so it cannot drift from the data.]

### C.1 The complete derived-rule registry

Every derived field, the INCI markers that trigger it, the authority for the rule and its grade. Grades are as defined in Section 4.8.2. Counts are products flagged, recomputed from the release identified in Section 4.13.

[TODO: For each rule this table must eventually also carry: the exclusion list, the rule version, whether all matches or only the first are retained in the evidence field, and the validation result for that rule. The registry below gives the marker set and the authority; the remaining columns are generated from the pipeline at release time.]

**Table C.1.** Every derived field, the INCI markers that trigger it, the authority for the rule and its grade. A regulation with an article number; B peer-reviewed result; C industry practice or a screening flag under a proposal or review, named as such; D a label defined for this project. B/D marks a rule whose general principle is published but whose operational form — the marker list, the number of levels, the keyword mapping — was defined here and is not itself validated. Marker lists are abbreviated where they exceed the column; the full lists and the exclusion lists are in the published script. Counts are products flagged, of 12,629.

| Field                                     | Markers matched in the INCI list                                                                                 | Authority for the rule                                                      | Gr. | Flagged |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----|---------|
| <i>Derived from the regulation itself</i> |                                                                                                                  |                                                                             |     |         |
| <b>nanomaterial</b>                       | the bracketed word ( <b>nano</b> )                                                                               | Art. 19(1)(g) declaration [40]                                              | A   | 101     |
| <b>solid_microplastic</b>                 | <b>nylon-<i>n</i></b> , polyethylene, polypropylene, PMMA, polyurethane, styrene/acrylates, PTFE                 | Reg. (EU) 2023/2055, <i>solid</i> particles [3]                             | A   | 589     |
| <b>synthetic_polymer</b>                  | the above plus carbomer, acrylates copolymer, polyquaternium, crosspolymers                                      | Same instrument; reported separately as these are swollen, not solid        | A   | 3,166   |
| <b>endocrine_concern</b>                  | butyl-/propylparaben, benzophenone-3, triclosan, BHA, cyclosiloxanes, resorcinol                                 | Under SCCS <i>review</i> : a screening flag, not an enacted restriction     | C   | 1,582   |
| <b>pfas</b>                               | <b>perfluor</b> , <b>polyfluor</b> , <b>fluoro</b> , PTFE                                                        | <i>Proposed</i> restriction; a broad <b>fluoro</b> match also over-captures | C   | 51      |
| <b>restricted_ingredients</b>             | exact match against the register                                                                                 | Annexes to Reg. (EC) 1223/2009 [38]                                         | A   | 9,613   |
| <b>lead_active_position</b>               | position of the first active in the list                                                                         | Descending-weight order, Art. 19(1)(g) [40]                                 | A   | 4,376   |
| <i>Derived from a published result</i>    |                                                                                                                  |                                                                             |     |         |
| <b>supports_barrier_repair</b>            | ceramide, cholesterol, phytosphingosine, sphingolipid, niacinamide, panthenol, centella, squalane, linoleic acid | Lipid mixtures B [70]; niacinamide [95]; the marker list is unvalidated     | B/D | 5,690   |
| <b>contains_ceramide</b>                  | <b>ceramide</b>                                                                                                  | Same                                                                        | B   | 1,064   |
| <b>feeds_malassezia</b>                   | polysorbates, sorbitan esters, C11–C24 fatty acids, PEG stearates/laurates/oleates, ferments                     | Lipid dependence [7]                                                        | B   | 7,093   |
| <b>use_time (night)</b>                   | retinol, retinyl, retinal, tretinoin, adapalene, benzoyl peroxide, hydroquinone                                  | Retinoid photostability [2]                                                 | B   | —       |

*continued overleaf*

| Field                                                    | Markers matched in the INCI list                                                                                                   | Authority for the rule                                       | Gr. | Flagged |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----|---------|
| may_cause_purging                                        | the above plus salicylic, glycolic, lactic, mandelic, azelaic acid                                                                 | Aneri et al. [13]                                            | B   | 2,510   |
| do_not_layer_with                                        | triggered by an ingredient family being present, not by a pair co-occurring                                                        | Principle B [14, 27]; the warning list is this project's     | B/D | 3,002   |
| routine_step                                             | product type mapped to an eleven-step sequence                                                                                     | Principle B [21, 27]; the eleven steps are this project's    | B/D | 12,629  |
| strength_level                                           | actives present, and their position                                                                                                | Principle B [43]; the three levels are this project's        | B/D | 11,802  |
| texture                                                  | product descriptor mapped to a sensory category                                                                                    | Vocabulary B [10, 48]; the keyword mapping is this project's | B/D | 7,432   |
| brand_parent_company                                     | brand mapped to owner                                                                                                              | Company filings                                              | B   | 3,109   |
| <i>Derived from industry practice, named as practice</i> |                                                                                                                                    |                                                              |     |         |
| contains_comedogenic                                     | isopropyl myristate/palmitate, coconut oil, lauric acid, octyl and ethylhexyl palmitate, laureth-4, SLS, cocoa butter (17 markers) | Rabbit-ear assay [44]; contested [6, 35]                     | C   | 1,991   |
| contains_aha                                             | glycolic, lactic, mandelic, malic, tartaric acid ( <i>citric acid excluded</i> )                                                   | Analysis-service vocabulary                                  | C   | 1,254   |
| contains_bha                                             | salicylic acid, betaine salicylate, capryloyl salicylic                                                                            | Same                                                         | C   | 1,026   |
| contains_pha                                             | gluconolactone, lactobionic acid                                                                                                   | Same                                                         | C   | 409     |
| contains_drying_alcohol                                  | alcohol denat., SD alcohol, ethanol, isopropyl alcohol ( <i>fatty alcohols excluded</i> )                                          | Same                                                         | C   | 826     |
| contains_fragrance                                       | parfum, aroma, and the named declarable allergens                                                                                  | Declarable list [39]                                         | C   | 5,047   |
| contains_animal_derived                                  | cera alba, lanolin, carmine, silk, collagen, keratin, snail secretion, propolis (34 markers)                                       | INCI Latin nomenclature                                      | C   | 1,641   |
| sunscreen_filter_type                                    | zinc oxide and titanium dioxide against the organic filters                                                                        | Mineral / chemical / hybrid convention                       | C   | 1,883   |
| contains_reef_harmful                                    | oxybenzone, octinoxate, octocrylene, 4-MBC                                                                                         | Jurisdictional bans                                          | C   | 747     |
| preservative_system                                      | six families from phenoxyethanol to the isothiazolinones                                                                           | Formulation convention                                       | C   | 11,802  |
| formula_base                                             | presence of <b>aqua</b> as first ingredient                                                                                        | Anhydrous formulas need no preservative                      | C   | 11,802  |
| moisturiser_type                                         | humectants against occlusives                                                                                                      | Same                                                         | C   | 11,802  |
| halal_candidate                                          | absence of the animal markers above                                                                                                | Derived under a second authority                             | C   | 9,415   |
| <i>Presentation labels, not defended as findings</i>     |                                                                                                                                    |                                                              |     |         |
| regional_style                                           | country of origin mapped to a tradition                                                                                            | Defined for this project                                     | D   | 5,186   |
| price_band_lebanon                                       | observed price mapped to a band                                                                                                    | Bands defined here; prices are observed                      | D   | 11,700  |

continued overleaf

| Field     | Markers matched in the INCI list          | Authority for the rule        | Gr. | Flagged |
|-----------|-------------------------------------------|-------------------------------|-----|---------|
| body_site | product type mapped to an anatomical term | Values aligned to UBERON [75] | D   | 12,629  |

## Chapter D

# Queries

[TODO: The SPARQL queries behind Chapter 8, and the SHACL shapes from Chapter 6, as listings.]



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